WORKING TITLE

Linguistic tells and business themes in the trademark record

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Abstract

All 13.99 million USPTO trademark case files are re-parsed into event-dated prosecution histories, and every filing is scored on how surprising its goods/services language is against what its industry filed in the five years before and the five years after. The average of the two surprises is the filing's atypicality; their difference is its lead, positive for wording the industry then moved toward. Among 3.10 million registrations, the most leading fifth are cancelled for non-use at the five-year gate 1.4 points more often than the most lagging fifth, within Nice class and registration year with drafting and counsel held ($t = 8.3$). The penalty holds under fifty global themes, five hundred, and fifty themes fitted per class; the protection attributed to atypical language does not, falling from 4.4 to 0.7 points under per-class themes, because a global model reads a theme imported from another class as atypical. The penalty was two to three times larger for marks filed in the late 1990s, reversed for 2000–2004 filings, and returned from 2008, with the swing inside four technology classes. Among one million owners debuting 2009–2018, the debut's language does not predict a priced round; SEC reporting and listing fall in lead. The firms holding most linked revenue described their first product in language their industry already used. Applicants who refile an abandoned mark rewrite toward the class's typical description, so every registered description is in part a conformed one. Code, panels, and the corpus build are public at `https://github.com/ericsilver/tm-vocabulary`.

1 Introduction

The pioneers are the ones with arrows in their backs. The saying is older than any dataset. It has been tested on named markets and naming choices (Golder & Tellis, 1993; Semadeni & Anderson, 2010), but not on a record where being early can be read directly from what every entrant wrote, because that record did not exist in usable form. It does now. Every US trademark application since 1884 is public, with its goods/services description, its owner, its counsel of record, and a dated prosecution history that records whether the mark registered, whether its owner proved five years later that the product was still in commerce, and whether it was renewed at ten. This paper re-parses all 13.99 million case files into that dated form and asks of each filing two questions about its language: how unusual was this description for its industry, and was it written in words the industry was moving toward or words it was leaving behind. The first is the filing's *atypicality*, the second its *lead*. The paper then follows the filing, and the firm behind it, through the gates the record provides.

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The findings build on one another, and each section is written to be read on its own.

Section 2 builds the measure. A description is reduced to the themes it draws on, fitted once over the whole corpus by a topic model and held fixed; its theme proportions are compared, by Kullback–Leibler divergence, to the average proportions of its own Nice class — the international classification of goods and services under which every application is filed, forty-five classes, a filing claiming one or several — in the five years before its filing date and the five years after. The average of the two divergences is atypicality; their difference is lead. The section states why words are not scored directly (a divergence on words counts the length of the filing), why surprise is measured within a class and never across classes, and what the measure does not see, which is how a filing combines its themes.

Section 3 follows every filing through registration and the first use-proof gate, on the largest sample the record allows. Registration rewards a middling atypicality and is indifferent to lead. The gate is not: among 3.10 million registrations, marks whose language most led their industry were cancelled for non-use 1.4 points more often than marks whose language most lagged it, within Nice class and registration year, with drafting and counsel held. The penalty is present in 33 of 44 classes and in both self-filed and represented marks, which fail at very different rates (69% against 47%) for reasons the section describes. It is not constant in time: it was two to three times larger for marks filed in the late 1990s, reversed for marks filed 2000–2004, and returned from 2008, with the whole swing inside four technology classes and, within them, in which themes each era's leading filings were drawn from. The section also shows what applicants do when an application is abandoned and refiled: they rewrite toward the class's typical description, from both tails, whether the owner or counsel drafts the second attempt, which means every registered description is in part a conformed one. It breaks out the internet as a theme and finds web-based filings in industrial goods classes failing the gate at the rate of software registrations rather than of their class. It closes with the reason the rest of the paper conditions on reaching the gate: registration is a launch indicator, nine in ten failures being the applicant's own inaction, and the gate is the first point at which the market's verdict reaches the record.

Section 4 asks whether the gate result depends on how the vocabulary was partitioned. It does not, for lead: under fifty global themes, five hundred global themes, and fifty themes fitted separately within each class, the penalty is +2.3, +2.6 and +1.9 points. It does, for atypicality: the protection the global scorings attribute to atypical language (4.3–4.4 points) falls to 0.7 under per-class themes, because what a global model calls atypical is largely a filing importing a theme from another class. The section measures that population directly: registrations carrying a theme new to their class fail slightly less than established ones, and more once their own atypicality is held.

Section 5 follows the owner rather than the mark, through a priced private round, SEC reporting, and an initial public offering. On one million owners debuting 2009–2018, funding does not vary with either axis of the debut's language; reporting and listing fall in lead and, with length held, rise in atypicality. The firms that came to hold most of the value (ten owners hold 18.5% of $19.2 trillion of linked peak revenue, five hundred hold 83%) described their first product in language their industry already used. Hand-curated vocabularies for the internet, artificial intelligence and blockchain show AI and blockchain debuts funded four to five times as often as the average debut, and listing about twice as often. Patenting and funding do not order each other in time. Neither axis of the language is a proxy for patenting.

Section 6 reports what changes under other choices: theme resolution, fitting noise, decayed reference windows, annual reference buckets, window width and mix, the absence of conditioning on grant, and the period. In each case the body's simpler formulation is stated beside the more complicated finding. One of them is a correction worth carrying in mind: decayed windows return a lead penalty about a seventh smaller than flat ones, with the same sign and precision.

What the record says, read together, is this. Being early with the language of a product costs at the use-proof gate, in three of every four product classes and under every partition of the vocabulary, and it costs more in the industries and the years in which the language was moving fastest. Being unusual is rewarded at the same gate, but the reward belongs mostly to filings that bring a theme in from elsewhere, and it is the examiner, not the market, that rewards middling conformity at registration. The firms that thrived were not the pioneers of language. They were written in the words their industries already used, and they are few.

2 Context, approach, and vocabulary

2.1 The record and the question

A US trademark application carries a description of the goods or services the mark will cover. The description is written to secure scope of protection: wide enough to cover the planned use, narrow enough to be granted. It is therefore a forced, dated disclosure of what a firm expected to sell. The USPTO publishes every application since 1884 with its prosecution history, and this paper re-parses all 13.99 million case files into dated events, so that each application can be followed from filing to registration, to the sworn proof of continued use required between the fifth and sixth year, and to cancellation or renewal.

The question is whether the language of the description, read against the language of the rest of the industry at the time, predicts what happened to the mark and to the firm behind it. Two properties of the language are measured. *Atypicality* is how unusual the description is for its industry. *Lead* is whether its language is ahead of or behind where the industry's language was moving. Sections 3–5 relate the two to registration, to survival of the use-proof gate, to venture funding and public listing. Section 6 reports what changes under other choices.

2.2 Two readers, one living backwards

Imagine two readers of a filing's goods/services description, each having seen a disjoint half of its industry. One has read only what the industry filed in the five years before it. The other, living backwards as Merlin was said to, has read only the five years after. Each is asked how surprising the wording is. The measurement is how much they disagree: a description that startles the first reader and not the second arrived before its industry's language moved that way.

Both windows are anchored on the filing's own date, the 1,826 days before it and the 1,826 days after, so no two filings made on different days share a reference, and filings made on the same day fall in neither of the other's windows. Inside a window every day counts the same. The measure asks whether wording is unusual against a period, not whether it is ahead of a trend inside that period; it has no momentum channel.

2.3 What is measured

The quantity is information content in Shannon's sense. Something that happens with probability p carries $-\log p$ of information when it happens: a common thing carries little, a rare thing a lot. Here the things are the themes a description draws on, and the probabilities are how often the filing's own industry drew on each of them in the reference window. A filing's score is the average information content of its themes under its industry's frequencies, weighted by how heavily it uses each, less the same average taken under the filing's own frequencies. It is zero for a description composed exactly as the industry composes its descriptions and grows as the filing puts weight where the industry puts little.

The apparatus is that of Murdock et al. (2017), developed on Darwin's reading notebooks, and Barron et al. (2018), who extended it to French Revolution parliamentary debates. They score a document against those before it and again against those after, calling the two quantities *novelty* and *transience* and their signed difference *resonance*. Their object was the life of ideas: how often an idea appeared, and how ideas stood in relation to one another, across one reading life or one assembly's debate. The object here is the composition of commercial offerings, which components a product or service is described as having, and how that composition shifts across an industry over time. The arithmetic carries over unchanged. The interpretation does not: their document has one author and the reference stream is that author's own reading, so a high value says something about a mind; here a filing is scored against other firms' filings, and its author is usually counsel drafting for scope of protection.

Themes, not topics. The clusters the model fits are called *themes* throughout. In the source literature and in the name of the method, latent Dirichlet allocation, the same objects are called topics, and a document's topic is what it is about. A goods/services description is not about anything in that sense; it lists components. A cluster of terms that co-occur across descriptions (*video, music, games, electronic, audio*) is a component that many offerings share, not a subject. Theme is the word for that. Topic is used here only in the name of the model.

2.4 From a description to two numbers

Counting words. A description is reduced to a bag of words: the text is lowercased, split into tokens of three or more letters, and counted, with adjacent pairs counted as tokens in their own right so that *computer software* registers alongside *computer* and *software*. Order is discarded. Take the filing for AMAZON.COM, lodged in the advertising and retail class on 18 April 1997:

> computerized on line search and ordering service featuring the wholesale and retail distribution of books, music, motion pictures, multimedia products and computer software in the form of printed books, audiocassettes, videocassettes, compact disks, floppy disks, CD ROMs, and direct digital transmission

Of its 45 distinct terms, 33 are in the theme model's vocabulary, among them *computerized*, *search*, *ordering*, *ordering service*, *retail*, *distribution*, *printed books* and *digital transmission*.

Grouping words into themes. Each filing is then re-described in a fixed, much smaller set of coordinates. Latent Dirichlet allocation is run once over a stratified sample of the whole corpus, all 45 classes together: it learns 50 clusters of words that tend to occur together and, for any filing, the proportions in which its words draw on them. Nothing supervises this; the clusters come from co-occurrence alone. The themes are shared across classes; what is specific to a class is the reference, the average theme proportions of that class's own filings in the window. The Amazon filing draws on six themes at more than a fiftieth of its weight: 0.41 on media and entertainment goods (*video, computer, electronic, music, games, audio*), 0.15 on computer and information services (*services, computer, software, information, data*), 0.13 on retail store services (*store, retail, store services, featuring*), 0.10 on printed matter (*paper, books, printed, cards, stationery*), 0.09 on downloadable and online software, and 0.07 on marketing of consumer goods. The online appendix lists every theme with its leading words and its weight in each class.[1]

[1] `https://aporia.institute/tm-vocabulary/online-appendix/themes_T50.html`

Fifty is coarse, roughly one theme per industry. Section 4 refits the model at 500 themes and separately at 50 themes within each class and repeats the gate analysis under each; Section 6 reports the resolution sweep and a second fit at the same resolution with a different seed. Fifty is kept in Sections 3 and 5 because it is the coarsest resolution at which every result holds, and because its lead contrast is the lowest of the resolutions tried.

Comparing against the two references. Write P for the filing's distribution over themes, and Q_{past} and Q_{future} for that distribution averaged over every same-class filing in the two windows, $[d_i - W, d_i)$ and $(d_i, d_i + W]$, with $W = 1{,}826$ days. How far P sits from a reference Q is the Kullback–Leibler divergence,

$$\mathrm{KL}(P \,\|\, Q) \;=\; \sum_k P_k \log \frac{P_k}{Q_k},$$

a sum over the 50 themes, zero when the two distributions agree and growing as the filing puts weight where the industry puts little. It is measured in nats, the natural-log unit. Write

$$K^- = \mathrm{KL}(P \,\|\, Q_{\text{past}}), \qquad K^+ = \mathrm{KL}(P \,\|\, Q_{\text{future}}),$$

so K^- is what the first reader feels and K^+ the second: the filing's past-facing and future-facing surprise.

Taking the difference. Amazon's filing is scored against 41,166 earlier filings and 146,532 later ones, giving $K^- = 1.286$ and $K^+ = 1.163$. The two axes used throughout are their average and their signed difference:

$$\underbrace{A \;=\; \tfrac{1}{2}\left(K^- + K^+\right)}_{\text{atypicality}}, \qquad \underbrace{L \;=\; K^- - K^+}_{\text{lead}}.$$

For Amazon, $A = 1.225$ and $L = +0.123$. Against its own class those say something the raw pair does not: its atypicality is at the 47th percentile, entirely ordinary, while its lead is at the 94th. The industry moved toward this description in the five years after it was filed and had not been writing that way in the five years before. Nothing about the wording was strange; it was early. L is signed, positive for a leading filing and negative for a lagging one, and the symbol $\Delta\mathrm{KL}$ is used interchangeably with it.

What "early" means is visible in the themes themselves. Of the six Amazon drew on, the share of class-035 filings drawing on each, in the five years before its filing and the five after:

Theme	Amazon's weight	Before (%)	After (%)
Retail store services	0.13	42.6	50.3
Computer and information services	0.15	58.9	59.9
Media and entertainment goods	0.41	9.9	11.3
Downloadable and online software	0.09	2.3	4.1
Printed matter	0.10	8.3	7.8
Marketing of consumer goods	0.07	10.5	9.3

A theme counts as drawn on when it carries at least a tenth of a filing's weight. The three themes that carry most of Amazon's description, retail, media goods, and the still-small online-software theme, all became more common in the class over the following five years, the last nearly doubling; the two minor themes it also touched became slightly less so. The filing is closer to the class's future composition than to its past one, and that, not any rare word, is its lead.

Differencing is what makes that separation possible. K^- and K^+ correlate at 0.988 on the software class, so entering both is close to entering one variable twice; rotating them into an average and a difference isolates the large common component from the small residual that carries the timing, and leaves the two regressors nearly uncorrelated (+0.036 against each other, where K^- alone would give +0.114). The cost is that L is a small residual of two large quantities, about a sixth of either level's spread, which Section 6 quantifies.

Table 1: The quantities, their sources, and the terms used here. *Lagging* and *leading* describe the two signs of L; *atypicality* is unsigned and says nothing about direction.

Quantity	Definition	Murdock et al.	Term used here
Surprise against the past	K^-	novelty	past-facing surprise
Surprise against the future	K^+	transience	future-facing surprise
Their average	$A = \frac{1}{2}(K^- + K^+)$	—	**atypicality**
Their signed difference	$L = K^- - K^+$	resonance	**lead** (leading / lagging)
Its magnitude	$\lvert L\rvert$	—	lead magnitude

Why the words are not scored directly. The themes are an extra step, and the words themselves could be compared to the reference: build the filing's distribution over the class vocabulary from the words it uses, do the same for every filing in the window, and take the divergence. The paper reports that scoring as a cross-check. It cannot be the primary one, because a divergence measured on words mostly counts the length of the filing. The divergence is an average surprise less a spread: for any two distributions,

$$\mathrm{KL}(P\,\|\,Q) \;=\; \underbrace{-\textstyle\sum_k P_k \log Q_k}_{\text{how unexpected these words are}} \;-\; \underbrace{\left(-\textstyle\sum_k P_k \log P_k\right)}_{\text{how spread out the filing is}} \;,$$

and the spread of a filing over words is bounded by how many distinct words it has: a description using k terms cannot spread over more than k of them, so its spread cannot exceed $\log k$, while the reference is spread over about 900 terms. A five-word description is charged the whole difference in width, roughly five nats, before any of its content is considered. Measured rather than derived: take 3,000 software-class filings of at least 150 words, score each against one fixed reference, then truncate the same filings to a random handful of words and score them again. Across a fiftyfold cut in length the surprise term moves by 0.016 nats and the spread term by 2.9. Synthetic filings whose words are drawn at random from the reference itself, maximally ordinary by construction, score 5.72 at three tokens and 1.65 at four hundred, tracking log(distinct terms) to within a fiftieth of a nat, and the average real filing scores marginally below that null at its own length. Term-scored atypicality correlates with the log of a filing's distinct term count at −0.68 in the software class.

Themes break the coupling because the 50 coordinates are always occupied. Under term scoring the effective number of coordinates a filing spreads over rises seventeenfold from the shortest descriptions to the longest; under themes it varies by less than a third, and not monotonically: a three-word fragment spreads across more themes than a long description, since the model cannot tell which one it belongs to. The width term therefore stops carrying length, and what is left is the part that was always about content. Theme-scored atypicality correlates with the log term count at −0.22 in the software class and −0.10 in advertising and retail, against −0.68 and −0.62 under terms.

What the themes see, and what they do not. The score compares a filing's theme proportions to the average proportions of its industry, so it reads which themes a filing draws on and how heavily. It does not, in any material way, read how a filing combines them. Taking each filing's two leading themes and asking how often that pair occurs together relative to the two themes' separate frequencies: how rare the themes are individually explains 70% of the variance in atypicality, and the unusualness of the pairing adds under two points on top of it. A filing assembled from ordinary parts in a genuinely new configuration is therefore close to invisible to this measure.

Combinations can be scored in their own right, at two levels, and the two levels behave differently. Both are computed on the software class, and both are reported net of the filing's own lead and atypicality, so each is the increment over what the construct already measures.

At the theme level, take every pair of themes a filing draws on (at a tenth of its weight or more) and measure how much more or less often each pair occurs together than the two themes' separate frequencies would predict, in the class's filings of the five years before and the five after. The negative log of that ratio is the pairing's information content; averaging over a filing's pairs, weighted by their masses, and differencing the two windows gives a *pair lead*, positive when the class moved toward the filing's pairings. Filings in the top fifth of pair lead fail the first use-proof gate 2.3 points more often than those in the bottom fifth ($t = 8.4$, unchanged by the controls), and pair lead correlates with lead itself at only -0.18. Being early with an arrangement of themes is penalised in its own right, at a rate comparable to being early with the themes. The rarity of a filing's pairings, by contrast, carries nothing once its atypicality is held (-0.3 points, $t = -1.1$).

At the word level the sign reverses. For each filing, count the share of its adjacent word pairs that are new to the class — the pair appearing fewer than five times in the class's prior five years — while both words are established there. This is recombination: familiar parts, new adjacency. Filings in the top fifth of that share fail the gate 6.1 points *less* often than those in the bottom fifth, and 2.8 points less with lead and atypicality held ($t = -10.4$). Word recombination and theme-pair lead are uncorrelated ($r = -0.01$): they are different facts about a filing. What the record shows, then, is not that combination is invisible and neutral. Arranging themes the industry had not yet arranged is punished like arriving early; coining phrases from the industry's own words is rewarded. The construct the paper uses sees neither, and the two run in opposite directions, so its blindness to combination is a bounded omission rather than a hidden bias in one direction. The sections that follow use lead and atypicality of the parts; these two measures bound what they leave out.

2.5 Within class, not across it

Goods/services descriptions are drafted to secure scope of protection, not to describe a product distinctively, and counsel reuse language from the USPTO's own Acceptable Identification of Goods and Services Manual, from house precedent, and from competitors' successful filings. Identical wording across unrelated firms is therefore common, and it carries through to the scores: 17% of scored registrations share a value with another filing in the same class and year. Within a class, the measure picks up departure from a shared drafting convention. Across classes it does not: a class where the Manual supplies dense standard language shows a compressed distribution of atypicality, and one where firms describe novel services in prose a wide one, for reasons unrelated to how innovative either industry is. Every estimate is therefore taken within class and year, and magnitudes compare across classes only as signs and orderings.

The vocabulary is positional rather than causal. A leading filing was ahead of its industry's language, which implies a trajectory without asserting the filing caused it. Atypicality is textual, and is not trademark law's distinctiveness doctrine, which governs registrability along the generic-to-fanciful spectrum; the two attach to different objects, one to the mark and one to the description of the goods it covers.

2.6 What the construct is not called

Novelty and *transience*, in the sense of Murdock et al. (2017) and Barron et al. (2018), are the two KL levels themselves: novelty is divergence from the past, transience divergence from the future. Atypicality as used here is their average, and so is neither of them alone; *novelty* in particular is declined because it names only the backward-looking half of a quantity that faces both ways. *Resonance* is their signed difference, and is arithmetically what this paper calls lead. The word is avoided because it names a mechanism — the field resonating with an author's contribution — that this design cannot observe. A filing can sit where its class was heading because it was imitated, because it and the class responded to the same outside event, or by coincidence; the measure does not separate these, and *resonance* would assert the first.

Innovation is a property of the product, not of the sentence describing it: two firms selling the same thing can draft very differently, and a genuinely new product can be described in boilerplate (Appendix E). The measure sees prose.

Radicalness (Dewar & Dutton, 1986; Henderson & Clark, 1990; Tushman & Anderson, 1986) is the closest neighbour, since KL is a departure measure rather than a newness count and the Amazon exhibit is a large departure built from unremarkable components. It is declined because it claims that existing competences are devalued, and nothing here licenses the step from unusual language to devalued competence.

Distinctiveness has two prior occupants. In trademark law it is the registrability doctrine, proven in part by the continued use this paper uses as an outcome. In strategic management, optimal distinctiveness (Zhao et al., 2017) predicts an inverted-U in conformity — a shape that appears here at registration but not at listing, so the term would imply a theory is being tested when it is not.

What *atypicality* says is narrower than any of them: this description is unusual for its class and year. Whether the thing described was new, radical, or good is not measured.

3 Registration and the first use-proof gate

This section uses the largest sample the record allows: every class-record filed 1995–2018 for the registration step, and every unique registration issued 2002–2018 for the first use-proof gate. It relates atypicality and lead to each step, reports how the two steps differ for self-filed and professionally filed applications, describes how applicants amend a description between attempts, and closes with the reason the sections that follow are confined to marks that reached the first gate.

A US trademark passes through a sequence of dated, high-stakes steps, and each step leaves a record. An application is filed with a goods/services description; an examiner reviews it, often issuing office actions the applicant must answer; if allowed and (for intent-to-use filings) a statement of use is filed, the mark registers. Registration starts a maintenance clock: in years five to six the owner must file a Section 8 declaration proving continued use in commerce, with a six-month grace period; around year ten a combined Section 8 declaration and Section 9 renewal is due, and every ten years thereafter. A mark that misses a use-proof gate is cancelled. Madrid Protocol registrations (§66(a) basis) file Section 71 declarations on the same schedule and are flagged separately throughout. Graham et al. (2013) describe the administrative data; the event-dated corpus built here adds the full prosecution history of each case, so a cancellation is assigned to a specific gate by its date rather than inferred from a status snapshot.

Table 2: The trademark funnel in this corpus. Registration counts are unique serials; gate outcomes are event-dated (a Section 8 cancellation at registration age 4.0–8.5 years). Source: the re-parsed USPTO application backfile, scored as in Section 2.

Stage	n	Share
Debut filings scored, 1995–2018 (owners' first filings)	3,589,068	
reached registration	2,066,600	57.6%
All unregistered class-records filed 1995–2018	3,798,318	
never filed statement of use		42.6%
stopped responding in prosecution		45.3%
removed adversarially		4.3%
Registrations 2002–2018, scored, unique serials	3,104,534	
failed the first use-proof gate (event-dated)		52.3%
Passed first gate, cohorts 2002–2013, terminal status known	1,129,724	
failed at the year-ten renewal		39.1%

The paper relates two properties of a filing's description — how unusual its language is for its industry, its *atypicality*, and whether that language is ahead of or behind where the industry's language was moving, its *lead* — to what happened to the filing afterwards. Section 2 builds the two properties. The outcomes are three steps in the sequence above, each read from the dated record and each bounded as follows.

Registration. Whether an application reaches registration at all, among filings made 1995–2018. Of the applications that do not register, roughly nine in ten die by the applicant's own inaction — never filing the statement of use, or going silent during prosecution — and only 4.3% are removed adversarially (Table 2), so registration is read as a launch indicator rather than an examiner's verdict.

Survival of the first use-proof gate. Whether a registration was cancelled for non-use at registration age 4.0 to 8.5 years, among registrations issued 2002–2018. This is where the market's verdict reaches the record: half of registered marks do not survive it, and failing it means the product the mark named is no longer in commerce. Every gate estimate is conditional on the mark having been granted, because registration is itself selected on language; what the unconditional composite looks like is shown once, in Appendix 6.8, and not otherwise used.

Reaching SEC reporting. Whether the owner of a first-time filing ever appears in SEC financial-statement records, a firm-level outcome used in Section 5.4 to ask whether either property of the text marks firms bound for public markets.

Outcomes are rates, so every estimate is a shift in a probability and is stated in percentage points against the base rate it shifts: +1.4pp on a 52.3% base means 53.7% against 52.3%, not 52.3% multiplied by 1.014. The basic comparison is the outcome rate of the fifth of filings with the highest lead minus that of the fifth with the lowest, with the fifths cut inside a single industry and year so that no difference between industries or between years enters; the regression tables add drafting and filer controls and cluster errors on the owner. Where a continuous version is given, it is the change in the outcome for a one-standard-deviation change in lead, in percentage points.

Counting differs by table. A filing may claim several Nice classes at once, so a *class-record* counts it once per class while a *registration* counts it once by serial number; a *debut filing* is an owner's first, one per owner. The funnel above counts debut filings, the gate regressions unique registrations, the per-class figures class-records.

Three windows bound the samples, each set by an institutional fact. Scoring runs 1995–2019, because a full five-year past reference does not exist before 1990 and a full forward reference does not exist after 2020; the analysis cut of 1995 is about two years more conservative than the measured burn-in requires (Appendix A). The gate cohorts end in 2018 because the cancellation record runs to April 2026 and the failure window closes at registration age 8.5 years, so 2018 is the last cohort whose window has essentially elapsed — the top of that range is fractionally censored, and a separate 2019 replication is reported below. They begin in 2002 because that is the conventional cut for this corpus, and that lower bound turns out to matter: carried back to the first scoreable cohorts the penalty is two to three times larger than it has ever been since, and Section 3.5 reports what the restricted window concealed.

Outcomes are read from the event-dated prosecution history, which attributes each cancellation to a specific gate and observes counsel and basis directly, and cross-checked against the USPTO status-code dictionary.[2]

3.1 Registration rewards a middling atypicality

Across 3.59 million debut filings (57.6% base rate), completion is an inverse-U in *atypicality*: it climbs from 46.6% at the least atypical end to about 58% in the middle and falls back to 53.4% at the most atypical (Figure 2, panel C). The same shape appears on the past-facing level alone, more mildly, running 54.0% at the bottom quintile to 58.4–59.1% in the middle and 57.7% at the top (Figure 1A).

The signed axis behaves differently. Completion on ΔKL falls to a trough of 51.9% just above the median and then rises steadily to 58.0% at the leading end. That trough is composition rather than timing: $|L|$ correlates with atypicality at $+0.31$, so filings near zero lead are disproportionately low-atypicality filings, which complete poorly for the reason above. Split by atypicality, the trough is absent in the upper three fifths and survives in the lowest. Reference windows built by calendar year put an inverse-U on this axis too, but that one is an artifact of annual bucketing (Appendix 6.5) and does not survive at daily resolution. The signed axis still enters a linear specification strongly (Table 10), but monotonically rather than through an interior peak: what curvature exists at registration belongs to how far a filing's language sits from its category, not to which side of it the filing sat.

The obvious objection is that unfamiliar language is simply refused more often, so the measure recovers nothing beyond familiarity. Completion falls at *both* ends, so the most category-typical filings also complete less than the middle, which familiarity cannot explain; and the curvature is specific to the unsigned axis, the signed axis showing no interior peak at all.

3.2 Self-filed and professionally filed applications

Counsel of record is observed on every filing. Represented applications are 73% of filings over 1995–2018, falling from 85% in 1995 to 68% in 2018 as online self-filing spread. The two populations differ at both steps. Represented applications reach registration 61.9% of the time and self-filed applications 45.6%. Among registrations, represented marks fail the first use-proof gate 47.5% of the time and self-filed marks 69.2%. The second gap is the larger, and it is not a drafting effect: the gate is proof of continued use, so a self-filed registration that fails it names a product that went out of commerce within six years, and self-filers are disproportionately new and small.

Self-filers also write differently. Their descriptions are shorter (a median of 21 terms against 28 in the software class, 17 against 26 in advertising and retail), sit closer to their class's usual

[2] 700 registered; 701–705 Section 8 accepted; 710 cancelled for Section 8 failure; 800 renewed; 900 expired.

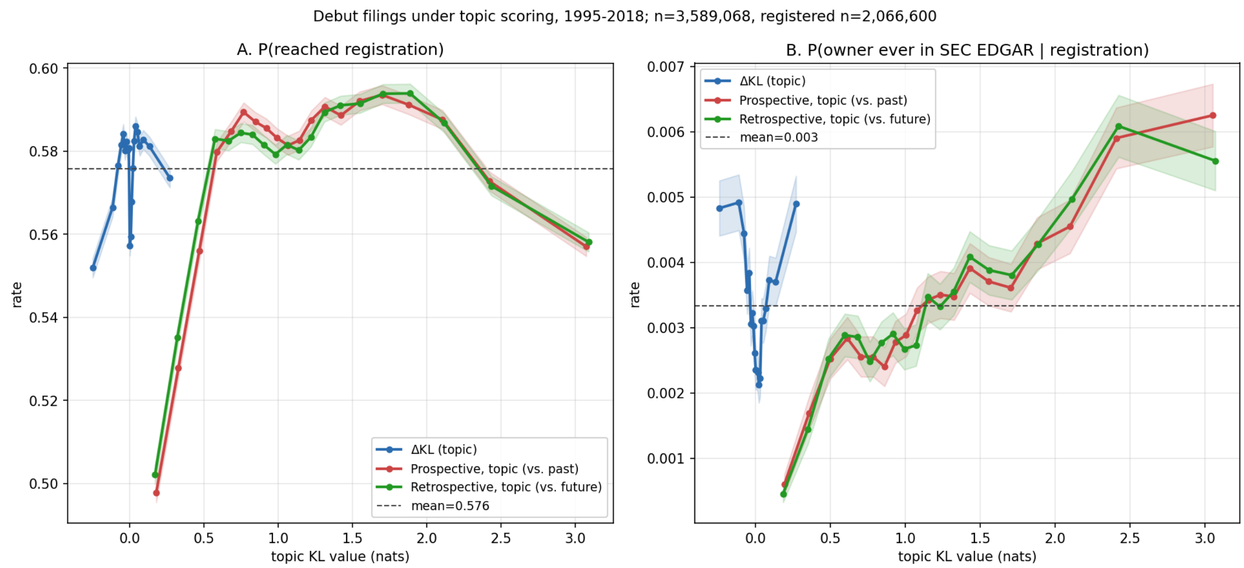

Figure 1: Outcomes for debut filings (owner's first-ever filing) across all 45 Nice classes under theme scoring, filing years 1995–2018, in quantile bins with 95% intervals. $n = 3{,}589{,}068$; registered subset $n = 2{,}066{,}600$. *A:* P(reached registration). *B:* P(owner ever in SEC EDGAR, the Commission's public filing system | registration); the structure in panel B is decomposed in Section 5.4.

language, and lean slightly further ahead of it. Counsel of record therefore moves both sides of every comparison below at once, which is why it is held in every regression and why the estimates are reported separately by stratum (Table 3). Held, the lead penalty is of the same order in both populations: +1.8pp among self-filed registrations and +1.2pp among represented ones.

3.3 How applications are amended

An application that is abandoned is usually the end of that attempt: among 2.28 million abandoned applications, 3.6% are followed within six years by the same owner filing the same mark again in the same class, and 1.5% by one that then registers. The refilings that do occur show what an applicant changes. Of the 42,115 abandoned applications whose owner later registered the same mark in the same class, 74% carry a different goods/services description the second time. Scoring each registration on the text it was first tried with and on the text it registered with gives the direction of the change. Both self-filers and represented owners rewrite toward the class's typical description: originals in the most atypical fifth lose 0.39 nats of atypicality, and originals in the most conventional fifth gain 0.16. Refilings that reuse the identical text move by less than 0.01 at every position, so the convergence is in the writing. Who drafts changes the extent but not the direction: owners who hire counsel for the second attempt rewrite most often (82%) and their descriptions lengthen by 16%; counsel rewriting their own earlier draft rewrite least often (72%) and shorten by 4% (Appendix C). Lead barely moves once the windows' shift with the later date is taken out.

Applicants therefore adjust descriptions to resemble what the class already files. A registered description is in part a conformed one, and the atypicality of registered marks is compressed relative to what applicants first wrote. Every estimate on registered marks is an estimate on conformed language. The refiled cases themselves move nothing: they are 1.1% of the gate sample, they fail

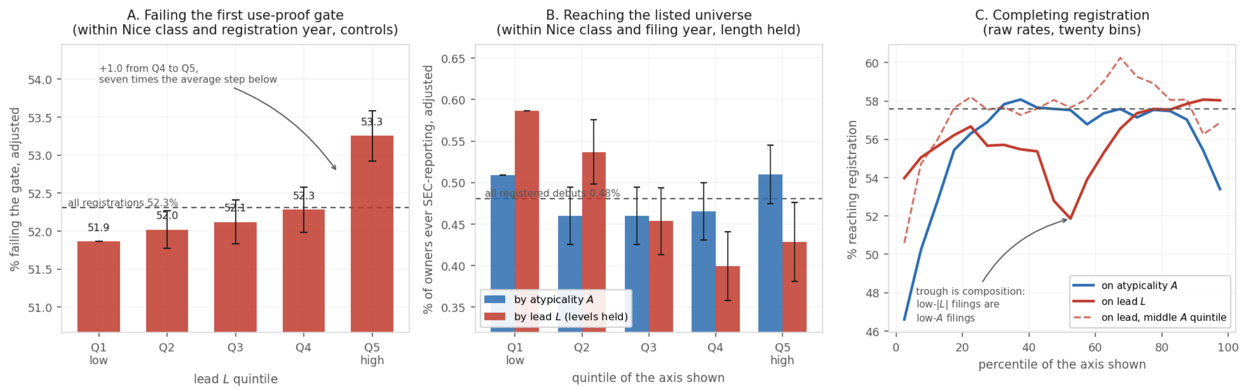

Figure 2: Quintile profiles for the three outcomes. *A:* the share of registrations failing the first use-proof gate, by lead quintile, adjusted within Nice class and registration year with the drafting and filer controls of Table 3; the dashed line is the rate for all registrations. The profile rises gently and monotonically through the fourth quintile and then steps up: about seven tenths of the top-to-bottom gap is the single step from the fourth quintile to the fifth, seven times the average step below it. A gradient is present, but the step at the top dominates it, and the text reports top-versus-bottom contrasts for that reason. *B:* the share of first-time filers whose owner ever reaches SEC reporting, by quintile of atypicality and of lead, adjusted within Nice class and filing year with description length held; flat in atypicality, and falling in lead through the fourth quintile once the levels are controlled. Whiskers in A and B are 95% intervals on the difference from the bottom quintile. *C:* raw registration rates for debut filings in twenty bins — an inverse-U on atypicality, and on lead a trough just above the median that is compositional, absent once atypicality is held in its middle fifth (dashed). Horizontal line at the pooled rate.

it slightly less often than other registrations (51.7% against 52.8%), and the within-class-and-year contrast is +2.25pp with them and without them.

3.4 Leading marks fail the first use-proof gate

Marks written in language their industry subsequently moved toward fail their first use-proof gate more often than marks written in language it was moving away from. Within Nice class and registration year, controlling for whether the filer drafted the mark themselves or hired counsel, and for the length of the description, marks whose language most leads their industry fail 1.4 points more often (53.3%) than those in the most lagging fifth (51.9%; $t = 8.3$, Table 3).

Among 4.12 million registration class-records 2002–2018, dated first-gate cancellation rises from 50.0% at the bottom theme-ΔKL quintile to 52.4% at the top, positive in 33 of 44 classes with sufficient volume (Figure 3). But a raw pooled contrast mixes the within-industry gradient with class and registration-year composition, treats multi-class registrations as independent observations, and ignores that gate outcomes correlate within owner. The primary estimates therefore come from a linear probability model on 3.10 million unique registrations, one row per serial, with ΔKL quintiles cut *within* Nice class and registration year. The specification adds class×registration-year fixed effects; controls for counsel of record, intent-to-use basis, log description length, log owner filing volume, owner domicile, and foreign filing basis;[3] and standard errors clustered on 1.27 million

[3] A US application is filed either on *use* — the mark is already in commerce — or on *intent to use*, which defers proof until after allowance. Foreign applicants may instead file on a home registration (§44(e)) or extend an international registration into the US under the Madrid Protocol (§66(a)); the latter maintain under §71 rather than §8. These are

normalized owners (Table 3).[4]

Table 3: First-gate failure and ΔKL: design-based estimates. LPM of event-dated first-gate failure on within-class×registration-year ΔKL quintiles (Q1 omitted), unique serials, registrations 2002–2018. Failure is a dated cancellation for non-use at registration age 4.0–8.5 years, under §8 or, for Madrid §66(a) registrations, §71. All specifications include class×registration-year fixed effects; controls are counsel of record, intent-to-use basis, log goods/services length, log owner filing count, owner domicile (US, CN), and foreign basis (§44(e), §66(a)). Base failure rate 52.3% on this sample.

Sample	Specification	Q5−Q1 (pp)	SE	n
All	FE only	+2.29	0.16	3,104,534
All	FE + controls	+1.39	0.17	3,104,534
Counsel-represented	FE + controls	+1.15	0.20	2,414,318
Counsel + US-domiciled	FE + controls	+1.60	0.22	1,888,352
Self-filed	FE + controls	+1.80	0.26	690,216
Excluding Madrid §66(a)	FE + controls	+1.82	0.18	2,924,427
Counsel + US, 2002–07	FE + controls	−0.25	0.33	551,989
Counsel + US, 2008–12	FE + controls	+2.00	0.42	551,924
Counsel + US, 2013–18	FE + controls	+2.65	0.25	784,439
All	continuous, pp per sd of lead	+0.58	0.05	3,104,534
Counsel + US	continuous, pp per sd of lead	+0.63	0.06	1,888,352

Standard errors replace significance stars: with n in the millions every coefficient has $p < 0.01$, so stars would mark each row identically. Errors are clustered on *normalized owner* — names uppercased, stripped of punctuation and of twenty legal suffixes, so ACME CORP. and ACME CORPORATION are one cluster. A single owner files many marks, drafts them in a house style, and abandons them together when the business fails; the within-owner correlation of gate survival is 0.38 pooled across classes (0.42 in the single-class checks where it was first measured), so the 1.27 million owner clusters, not the filings, carry the information. The Madrid row drops §66(a) registrations rather than absorbing them in a control, since they maintain under a different statute; the estimate is larger without them.

The gradient is monotone within cells. Whether the filing was drafted by counsel or by the owner is held because it moves both sides of the comparison at once: self-filers fail the gate at 69.6% against 46.8% for represented owners, and they write shorter descriptions that sit closer to their class's usual language yet lean slightly further ahead of it, so without the control, part of any lead gradient is a self-filing gradient. With it, the penalty is of the same order in both strata (+1.8pp self-filed, +1.2pp counselled). The full control set cuts the magnitude by about a third without absorbing it, and the estimate is if anything *largest* in the cleanest stratum — counsel-represented, US-domiciled owners — so it is not carried by the self-filed or foreign mass-filing segments whose gate failure rates are high for their own reasons.[5] About forty percent of the raw pooled contrast

statutory routes to the same registration, and they differ sharply in who files and how often the mark is abandoned, which is why each enters as a control.

[4]Ties are common. Within the class×registration-year cell the quintiles are cut in, 17.1% of scored registrations share a ΔKL value with at least one other filing, and the largest tie group runs to 53. Anchoring the reference window on the filing date does not remove them, because filings that share a class, a filing date and a goods/services description — an owner registering a family of marks at once, on Manual language — have identical distributions scored against identical references. Cut points are fixed by sorting on the score and then on the serial number, which is reproducible but systematic: serials run in filing order. It makes no difference. Sorting the ties the opposite way moves the top-versus-bottom contrast by 0.001pp, and five pseudorandom orderings move it by at most 0.002pp, against a contrast of 2.287pp — a tenth of one percent of the estimate.

[5]The level effects on those covariates are large in their own right: counsel −19.5pp and Madrid §66(a) basis +10.3pp.

is composition: +1.4pp design-based against +2.5pp raw.

Within the 2002–2018 window the penalty looks modern: indistinguishable from zero for the 2002–2005 cohorts, turning positive in 2006, and running +1.8 to +5.6pp for every cohort from 2008 onward (Figure 4). That reading does not survive carrying the series back: the penalty was two to three times larger for the earliest scoreable cohorts than for any since, which makes the 2002–2005 quiet a trough between two episodes rather than a starting point (Section 3.5).

The gate is close to a fixed appointment rather than a hazard spread over years: of the 1.35 million cancellations recorded for fully observed cohorts, 17 post before registration age 6.5, 96.1% post between 6.5 and 7.0, and the rest after seven. Since the cancellation record ends on 2 April 2026, a cohort is observable only once it has reached seven years, which makes 2019 the last one with usable gate data and leaves 2020 and later with no observed gate at all. Measured on the 6.5–7.0 band, fully observed for every cohort it covers, the penalty runs between +2.1 and +5.6pp with no trend across the decade and overlapping intervals at its two highest points. The 2019 reading rests on the 78,408 registrations issued early enough in that year to reach seven years inside the record, so it is a first-quarter subsample rather than a cohort.

The outcome is an indicator on a window, so a mark that fails outside it counts as a survivor. If leading marks failed at the same rate as lagging ones but sooner, that alone would put them inside the window more often and the measure would report a difference in rates where there was only a difference in timing. It is not. On the 3.15 million registrations whose ninth anniversary is inside the record — for which failure is settled and no window is needed — the contrast is +2.71pp within class and registration year, against +2.65pp on the paper's 4.0–8.5 window and +2.65pp on the narrow band. Conditional on failing, the probability of failing late carries a contrast of −0.01pp ($t = -0.2$) once class and registration year are held. The pooled version of that contrast is +0.27pp and is composition.

The filer population changed over the same period. Within counsel-represented US-domiciled owners, and with the full control set, the earliest cohort third is slightly negative (−0.25pp, SE 0.33) while the two later thirds are clearly positive (+2.00 and +2.65pp, Table 3). The gradient is therefore not an artefact of the post-2015 surge in self-filed and foreign applications.

3.5 The penalty is not modern, and it is not general

Scores exist from 1995 on, which makes registration cohorts from 1996 evaluable, and their gate outcomes have been settled for two decades. Carried back, the contrast runs +6.9pp for the 1996 cohort and rises to +10.1pp for 1999 — two to three times anything observed since — then collapses. The 2002–2005 cohorts the paper had been reading as a quiet baseline are a trough between two episodes, and one of them is the larger.

Scores begin in 1995, so a 1996-cohort registration must have issued within a year of filing; early cohorts are selected on prosecution speed, and speed plausibly travels with drafting in standard language. Holding pendency in a common one-to-three-year band leaves the series unchanged. Indexing by filing year instead, so a cohort contains every prosecution speed, gives +7.9 to +9.6pp for filings 1995–1999 and −0.95pp for filings made in 2000 — a ten-point swing in a single filing year.

Splitting the four classes that carry most technology filings — 009, 035, 038 and 042 — against the other forty-one separates a large moving component from a small steady one (Figure 5):

Filing years	Technology classes	All others
1995–1999	+14.18 (0.36)	+4.48 (0.27)
2000–2004	−1.37 (0.35)	+1.21 (0.24)
2005–2007	+2.04 (0.43)	+1.99 (0.28)
2008–2014	+9.02 (0.25)	+1.32 (0.17)

Outside technology the penalty is a flat +1 to +4pp in every era, including the one in which the pooled series reads as zero. Inside technology it swings by fifteen points: strongly positive for filings made into the late 1990s, negative for those made in the five years after, and positive again from 2008. The pooled null of 2002–2005 is a technology reversal partly offsetting a non-technology positive, not an absence.

What this design cannot say is which cycle the swing belongs to. Marks filed 1995–1999 reach their gate in 2001–2007 and marks filed 2000–2004 reach theirs in 2006–2013, so a story in which unusual language is punished because it was written during an investment boom and a story in which it is punished because the gate falls during a bust both fit these numbers.

3.6 Three of the four classes reverse, and the leading language changes theme

The four-class split pools industries that do not move together. Taking them one at a time, and then assigning quintiles inside progressively finer cells (Table 4):

Table 4: The leading gate penalty inside technology, by class and by cell. Top-minus-bottom lead-quintile contrast in first-gate failure (pp, SE in parentheses), settled registrations, by filing era. The upper block assigns quintiles within each class and era. The lower block pools the four classes and assigns quintiles within the era (raw, as in the text table above), within class×registration-cohort cells, and within theme×class×registration-year cells, where a filing's theme is the one its description loads most heavily on under the production $T = 50$ model. n = 1,248,591.

Class	1995-1999	2000-2004	2005-2007	2008-2014
009 Electrical, software	+18.04 (0.52)	−5.32 (0.51)	+0.25 (0.62)	+11.67 (0.37)
035 Business services	+4.59 (0.69)	−9.70 (0.56)	+2.02 (0.63)	+1.66 (0.36)
038 Telecommunications	+9.91 (1.40)	−4.24 (1.29)	+2.60 (1.48)	+3.42 (0.94)
042 Scientific, computer services	+17.11 (0.57)	+6.70 (0.62)	+8.42 (0.87)	+12.63 (0.47)
Four classes pooled, raw	+14.49 (0.33)	−1.81 (0.31)	+2.56 (0.38)	+9.16 (0.22)
within class×cohort	+14.27 (0.33)	−3.63 (0.31)	+2.73 (0.38)	+7.24 (0.22)
within theme×class×cohort	+4.78 (0.33)	−1.26 (0.31)	+2.56 (0.38)	+4.51 (0.22)

Electrical and software goods (009) and telecommunications (038) follow the pooled pattern. Business services (035), the class that holds online retail, reverses hardest: −9.7pp for filings made 2000–2004 against +4.6pp in the five years before. Scientific and computer services (042) never reverses. Its contrast is +6.7pp in the era in which the other three are negative and between +8 and +17pp in every other, and by registration cohort it never falls below +3pp (Figure 6). The reversal is an event in goods and in business services; the class that sells technology as a service sat it out.

Assigning quintiles within class and registration year rather than within the era pool changes little. Assigning them within theme, class and registration year changes a great deal: the late-1990s contrast falls from +14.5 to +4.8pp, the post-2008 contrast from +9.2 to +4.5pp, and the

2000–2004 reversal from −3.6 to −1.3pp. Two thirds of the first and half of the second are between themes rather than within them. The leading fifth of each era was drawn from particular themes, and those themes carried their own failure rates; the penalty for being leading *inside* a theme is a steadier +2.5 to +5pp, the same order as the penalty outside technology.

Which themes is a matter of record (Table 5). In 1995–1999 the computer, software, information and data services theme held 38% of technology filings and 75% of the leading fifth, and its registrations from those filing years failed at 64.9% against 47.5–59.0% in the other large themes. That one theme is most of the late-1990s penalty. In 2000–2004 it held 12% of the leading fifth. The leading language had moved to video, music and games on electronic media (31% of the leading fifth against 12% of filings) and the internet-services language had become lagging — wording the class was leaving — and inside several themes the lagging fifth now failed more than the leading one: −3.6pp in electronic media, −5.5 in retail store services, −6.9 in apparatus and control systems. From 2008 the leading fifth over-represents downloadable and online software by four to one (15% against 4%), the vocabulary of the application store; those registrations fail at 67.2% over the whole period, the highest base rate of any theme, and their within-theme lead contrast pooled across cohorts, +8.4pp, is the largest of any theme (the table's 2008–2014 cell, cut within that era alone, is +3.3pp). The computer-services theme's own contrast, meanwhile, is positive in every era (+3.6 to +5.9pp) and never reverses.

Table 5: The leading gate penalty by theme within the four technology classes. Themes with at least 20,000 settled registrations, labelled by their five highest-probability terms; quintiles assigned within theme and era. Cells with fewer than 3,000 registrations are blank.

Theme (top words)	n	1995-1999	2000-2004	2005-2007	2008-2014
services, computer, software, information, data	451,253	+3.61 (0.52)	+3.56 (0.49)	+4.18 (0.64)	+5.92 (0.37)
video, computer, electronic, music, games	162,290	+2.53 (0.97)	−3.62 (0.89)	−2.85 (1.01)	+3.36 (0.60)
services, featuring, store, retail, store services	131,107	+8.39 (1.09)	−5.49 (1.01)	+2.92 (1.15)	+0.98 (0.66)
services, field, training, estate, educational	95,602	+7.82 (1.09)	+0.99 (1.12)	+6.02 (1.43)	+2.05 (0.83)
apparatus, systems, devices, electronic, control	84,121	+4.80 (1.27)	−6.92 (1.19)	−0.58 (1.41)	+3.65 (0.88)
design, field, systems, development, services	32,517	+10.40 (2.03)	+4.67 (1.82)	+1.50 (2.27)	−2.27 (1.45)
software, online, downloadable, non-downloadable, virtual	29,472	—	—	—	+3.30 (1.05)
bags, clothing, shirts, jackets, leather	28,421	+1.28 (2.59)	−1.55 (2.39)	−4.66 (2.40)	+6.06 (1.36)
research, medical, scientific, testing, purposes	25,897	+12.65 (2.66)	+12.09 (2.03)	−3.82 (2.59)	+5.63 (1.51)
services, health, field, care, information	22,880	+4.94 (2.04)	+1.39 (2.36)	—	+3.50 (1.72)

So the swing has two layers. Across the whole period, a leading filing is a few points more likely to fail than a lagging one in the same theme, class and cohort, in technology as elsewhere. On top of that, each era's leading vocabulary was concentrated in one or two themes whose marks failed at their own rate: internet services in the 1990s, application software after 2008. In the trough between, the language that had been leading became lagging, and the marks still written in it failed with it. The design-based estimates in Table 3 hold class and cohort fixed but not theme, so they carry both layers; a reader who wants only the within-theme penalty should take the lower block of Table 4 as the guide to its size.

Levels and gradients separate cleanly across filer strata. Self-filers fail the gate far more often than represented owners (69.6% versus 46.8%), and intent-to-use registrations more often than use-based ones (53.1% versus 50.2%), but the ΔKL gradient itself is similar across these strata once class, cohort, and drafting covariates are held fixed (+1.8pp self versus +1.2pp counseled in Table 3). Per class, the largest penalties sit in tobacco (+26pp, the vaping era), hand tools (+10pp), processed foods (+10pp) and medical apparatus (+9pp), with negative lifts in beverages, transport and construction. These are the extremes of a 44-class screen of raw contrasts: the largest of many noisy estimates sit away from zero by selection, so the named classes locate where the penalty concentrates and their magnitudes should be read with that in mind.

The risk persists at the year-ten renewal, and on the design-based estimates does not attenuate. Table 10 models survival, so its signs run opposite to the failure convention used above: lead enters at −0.49pp on passing the first gate and −0.63pp on renewing at the second — the same penalty, stated on the other side. What attenuates is the raw single-class contrast. In the software diagnostic class, where the event-dated first-gate lift on the 2010–2013 cohorts reaches +12.1pp, the conditional lift at the second gate falls to +7.6pp. Pooled across classes the second-gate profile is not a slope at all. Fitted over 872,079 first-gate survivors with a resolved year-ten status, a straight line through renewal against lead explains 1% of the variation across forty bins and a V explains 89% (ΔAIC = 83, the largest margin anywhere in this paper). The fitted shape is an inverted V: renewal is highest just below the centre of the lead axis (64.2% near $z = -0.4$) and lowest at both tails, 45.5% at the most lagging bin — the lowest anywhere on the axis — and 50.6% at the most leading. The −0.63pp linear coefficient reported for that row is therefore close to meaningless as a description, though it is not wrong about the direction of the leading tail. Among first-gate survivors, the most lagging renew least of all. Most of the difference between leading and lagging resolves at first proof of use; what remains at renewal is a penalty on both extremes of the lead axis, the lagging one the larger.

A competing reading is that the gate measures disengagement rather than products: an owner who has lost interest both answers the examiner slowly and lets the registration lapse. The prosecution record tests this through the gap between the first non-final office action and the response to it.

Response speed does not predict the gate. Across the 1.22 million scored registrations with a usable office-action/response pair, gate passage varies between 46.1% and 48.1% across response-speed quintiles, from a median of seven days to 183 — flat, and if anything the quickest responders pass slightly *less*. The lead penalty is present in both halves, stated here on passing, as the paragraph's outcome is: −3.48pp (±0.40) among faster-than-median responders and −2.36pp (±0.40) among slower, against −2.94pp pooled. The restriction drops 61% of the sample, disproportionately applications that never drew a refusal, so this speaks to the prosecuted subset.

Mixing the past and future reference windows identifies which kind of novelty the gate punishes. Scored against a long past and a short future, so that lead measures a break with what the class had established, the top-versus-bottom contrast is +3.0pp; scored against a short past and a long future, so that it measures resemblance to where the class was heading, it is +1.65pp. The symmetric five-year measure sits between, and the penalty rises monotonically along the mix. The gate punishes breaking with the past more than it punishes anticipating the future, but it punishes both (Appendix A).

3.7 Internet as a theme: web-based filings against their class

A filing is internet-bearing when its goods/services text names the internet, the web, a website, online or on-line provision, or electronic commerce. The rule reads the text rather than the class, so a web retailer filing in apparel is caught and a semiconductor maker filing in class 009 is not. Internet-bearing filings are 16.9% of scored registrations overall: 58% in telecommunications, 39–41% in advertising and retail, scientific and technical services, and legal and personal services, under 5% in most goods classes. Table 6 gives, for every class, the registration rate and the first-gate failure rate of internet-bearing filings against the rest of the class; Figure 7 draws the gate column.

Three patterns are in the table. In the industrial goods classes, internet-bearing registrations fail the gate far more often than their class: chemicals 46.1% against 39.1%, metal goods 44.5% against 39.1%, machinery 50.9% against 38.7%, rubber and plastics 44.1% against 38.5%, vehicles 49.6% against 45.1%, medical apparatus 51.7% against 45.4%. A web-based business registering in an engineering class fails at the rate of a software registration (51.4%), not at the rate of its class. In the consumer goods classes the sign reverses: internet-bearing registrations in clothing fail 52.4% against 58.7% for the class, in carpets 43.7% against 46.9%, in household utensils 44.5% against 50.5%, in paper and printed goods 47.9% against 50.7%. Online sellers of consumer goods outlast the class's other registrants. In every service class the internet-bearing filings fail more often, by one to eleven points, and most in legal and personal services (60.1% against 48.7%) and advertising and retail (57.9% against 52.0%). Pooled, internet-bearing registrations fail at 57.9% against 51.8% in the technology classes, 54.2% against 49.9% in the other service classes, and 50.1% against 50.1% in goods classes, where the industrial and consumer differences cancel.

Registration runs the other way in goods. Internet-bearing applications in goods classes reach registration 59.5% of the time against 56.0% for the rest, and in the technology and service classes slightly less often than the rest. The examiner's step and the market's step do not agree about these filings: an internet description in a goods class is a little more likely to be granted and, in the industrial classes, a good deal more likely to be cancelled for non-use six years later.

The penalty has a common shape in event time. Dating the internet's arrival in each class as the first filing year in which the pattern reaches 2% of the class's filings (1994–1995 in the technology and transport classes, 1999–2000 in apparel and chemicals, 2008 in machinery), and aligning each class's per-cohort gap on years since its own arrival: the gap between internet-bearing registrations and the rest starts near +7 points in the first observable cohorts, declines to about zero nine to eleven years after arrival, and settles between +1 and +4 (Figure 8). Across the eight classes observable in both phases, the mean gap is +4.4 points in years zero to four and +1.6 from year ten, shrinking in six of the eight. Calendar time disguises this: transport, whose arrival was 1995, peaks at +16 points in its 2000 cohort and declines; machinery, whose arrival was 2008, peaks at +17 points in its 2012–2015 cohorts — its own early years. The class's clock starts when the vocabulary arrives, not when the internet did. Clothing is the standing exception: its internet-bearing registrations outlast the class in every cohort after 2004, consistent with online sellers of goods surviving where early web-services ventures did not.

The classification system has faced this question and twice declined to make the internet a class. Before 2002, internet and computer services accumulated in class 42, whose heading then ended "services that cannot be classified in other classes"; the Nice Union's Committee of Experts restructured it in the eighth edition, effective 1 January 2002, by narrowing 42 to scientific and technological services and moving the rest into three new classes (43–45), organised by the kind of service rather than by the technology.[6] The question returned with virtual goods: the twelfth

[6] Nice Classification, 8th edition (WIPO, 2002); the USPTO's guidance describes the class-42 restructuring. The eighth edition took effect in the middle of the era swing of Section 3.5.

edition, effective 1 January 2023, classifies downloadable virtual goods and NFT-authenticated files in class 9 and distributes metaverse-related services across the existing service classes, again declining a new class.[7] The pattern in Table 6 bears on that choice. Internet-bearing filings in a goods class survive or fail at rates closer to the technology classes than to their own class, so for the outcome this paper measures, the technology overlay carries more information than the class assignment it is filed under. A classification built for scope of protection has no reason to track survival; but a reader using Nice classes as industry controls, as this paper and the literature do, should know that the internet cuts across them.

3.8 Why the rest of the paper conditions on reaching the first gate

Registration is not a neutral filter on language, so every estimate below conditions on it. Completion is an inverse-U in atypicality: applicants whose language sits far from their category in either direction follow through least often, and the penalty is larger for the conventional tail than the unusual one. Pooling granted and abandoned filings therefore mixes what examination does to a description with what the market does to the product, and the resulting gradients are attenuated and in places reversed (Appendix 6.8).

Conditioning would be the wrong move if abandonment were usually a pause rather than an ending — if firms reworked refused applications and refiled them, the same commercial intent would be counted once as a failure and again as a success. It is not. Among 2.28 million abandoned applications, only 3.6% are followed within six years by the same owner filing the same mark again in the same class, and only 1.5% by one that then registers. Crediting every successful refiling as an eventual registration moves the inverse-U in atypicality from a depth of +2.91pp to +2.85pp. Abandonment is overwhelmingly terminal, and what the registration step selects on is therefore a property of the filing rather than an artifact of counting.

The refilings that do occur were shown in Section 3.3 to be rewritten toward the class's typical description, so the atypicality of registered marks is compressed relative to what applicants first wrote, and every estimate on registered marks is an estimate on conformed language. For the 42,115 refiled cases the first text is observed; for an applicant who conformed before ever filing it is not, and that part of the compression cannot be bounded. The refiled cases themselves move nothing: they are 1.1% of the gate sample, they fail it slightly less often than other registrations (51.7% against 52.8%), and the within-class-and-year contrast is +2.25pp with them and without them. Which of the two texts predicts their gate cannot be told on 34,551 cases (+0.9pp on the rewrite, +0.7pp on the original, both with standard errors near 0.8pp).

What examination rewards is a middling specificity — concrete enough to survive a definiteness objection, conventional enough to avoid a likelihood-of-confusion refusal. Completion runs from 46.6% at the least atypical end to about 58% in the middle and back to 53.4% at the most atypical; Appendix B gives the profile on each axis and why familiarity alone cannot produce it. The rest of this section is about what happens to marks after the office has granted them.

Sections 4 and 5 are therefore confined to registrations, and within them to the first use-proof gate and to what follows it. A mark that never registered left the record by the applicant's inaction in nine cases of ten, before any product met a customer; a mark that registered and then failed the gate names a product that was in commerce and is not. The second is the outcome the paper is about. Appendix 6.8 shows what the pooled analysis looks like without the condition: the registration step's shape is stamped onto every later outcome, and the gradients the conditioned analysis finds are attenuated and in places reversed.

[7] Nice Classification, 12th edition (WIPO, 2023).

4 The year-five gate under three theme resolutions

Section 3 scored every filing on fifty themes fitted once over the whole corpus, with each class supplying its own reference. That is one of three ways to partition the vocabulary, and the gate result should not depend on which. This section repeats the first-gate analysis on the identical registrations under three scorings:

- *global, 50 themes*: one model over all 45 classes, references per class (the production scoring of Section 3);
- *global, 500 themes*: the same construction at ten times the resolution, so a theme is closer to a product category than to an industry;
- *per class, 50 themes*: a separate 50-theme model fitted on each class's own filings, so the partition is tuned to the class and nothing is shared across classes.

The third differs from the first in what a theme is, not in where the reference comes from; in all three, surprise is measured against the filing's own class. The second changes only the resolution. All three are scored on per-filing windows of 1,826 days on each side, and the gate outcome is the same event-dated cancellation for non-use.

4.1 Three scorings, one gate

Table 7 gives the top-minus-bottom quintile contrasts on the 3.10 million registrations 2002–2018 that all three scorings score, with quintiles cut within class and registration year.

The lead penalty is a property of the filing, not of the partition. It is +2.29pp under the production scoring, +2.61pp at 500 themes, and +1.94pp under per-class themes, each with a standard error of 0.09, and positive in 28 of 44 classes under all three at once. The scorings agree only moderately filing by filing: lead correlates at 0.64 between the two global resolutions and at 0.52 between the global and the per-class fifty, so a third of the variance in any one scoring's lead is particular to that scoring, and the penalty survives the disagreement. The standard errors in this section's raw contrasts are two-sample binomial and unclustered; outcomes correlate within owner at 0.38 (Section 3.4), so the t-statistics overstate precision, by roughly a factor of two on single-class cells. The pooled contrasts are far from any such margin; the marginal per-class and per-episode cells below are not, and are read as locating rather than establishing.

Atypicality behaves differently. Under the two global scorings the most atypical fifth of registrations fails the gate 4.3–4.4 points less often than the least atypical; under per-class themes the gap is 0.7 points. The two global scorings' atypicality correlates at 0.79; global and per-class atypicality correlate at 0.39. A global model places every class in one set of coordinates, so a filing that imports a theme the class rarely uses and other classes use often registers as atypical for its class. A model fitted on the class alone re-partitions the class's own vocabulary, and the imported theme becomes a local theme like any other. What the global scoring calls atypical is, in large part, drawing on a theme from elsewhere. Section 4.2 measures that population directly.

4.2 Themes new to the corpus and themes new to a class

A theme can be new in two senses. It can be new to the record: no class was filing it until recently. Or it can be established elsewhere and only now arriving in this class. The 500-theme model is fine enough to separate the two. For each theme and each class, the first sustained year is the first year by which the theme has been the dominant theme of at least one hundred filings in that class; the corpus-wide first sustained year is the earliest of those across classes. Each registration class-record in the gate sample is assigned its dominant theme and classified at its filing date as new to the corpus (filed before the third year after the theme's first sustained year anywhere, including filings made before the theme was sustained at all), new to the class (not new to the corpus, but filed before the third year after the theme's first sustained year in this class), or established.

Of 4.12 million registration class-records 2002–2018, 3.97 million carry an established theme, 151,012 a theme new to their class, and 779 a theme new to the corpus. The last group is too small to support a claim: almost every theme a 500-theme model can resolve had been established in some class before 2002, and the 779 that had not are scattered across small, late themes. The comparison the record supports is between established themes and themes new to the class.

Registrations carrying a theme new to their class fail the gate at 48.3% against 51.6% for established themes, and 0.43 points less often within class and registration year (SE 0.13). Holding the filing's own lead and atypicality as well reverses the sign: net of them, the new-to-class registrations fail 1.06 points more often. The two statements describe one population. A theme arrives in a class in filings that are atypical for that class, and atypical filings survive the gate; that is the −0.43. Among filings of equal atypicality, the ones carrying the arriving theme fail more; that is the +1.06. Importing a theme is one way to be atypical, and it is a more exposed way than the others.

4.3 Effective novelty: themes that surged

Since truly new themes are rare, the observable event is a surge: an established theme's share doubling, over its own three prior years, to a level that matters. Measured against the whole corpus at a 1% share, five themes ever surge, and one episode is large enough to read: the online-and-downloadable-software theme doubled to 2.4% of all filings in 1999. The 6,339 registrations whose dominant theme it was during 1999–2001 failed the first gate at 63.7%, 8.6 points above their class and year and 8.5 net of their own lead and atypicality. Within the episode, the most leading fifth failed no more than the most lagging (−0.2pp, SE 1.9). Everyone in the wave failed together, whether they were early to its language or late.

Measured within a class at the same thresholds, surges are common: 962 episodes of a theme doubling to at least 1% of its class, holding 32,418 gate registrations. Pooled, being in one of these smaller waves costs little: in-surge registrations fail 0.65 points more than their class and year, 1.27 net of lead and atypicality (1.33 and 1.70 at the 2% threshold). The episodes differ in sign among themselves — the education-services surge in advertising and retail after 2002 failed 13.7 points *less* than its class, the charitable-fundraising surge in finance after 2011 failed 4.0 points more — and of the few episodes large enough to carry their own lead contrast, one is clearly positive (the education-services surge itself, +5.8pp, $t = 2.2$) and the rest are indistinguishable from zero.

Three further facts about the waves, on the 61 episodes with at least 300 joining registrations over a six-year entry window — 64,105 of the 86,447 registrations that join any episode. First, within a wave, entry order does not matter: demeaned within episode, joiners in the surge year fail −0.8 points relative to their wave's mean, joiners five years in +0.0, with no gradient between (each SE about 0.5). Second, among class-scale waves, larger is not deadlier: excess failure is +2.7 points in the smallest tercile of waves, +3.9 in the middle, −0.1 in the largest, and correlates with wave size at −0.22. The eight-point excess of the corpus-scale software wave is not the top of a dose-response within classes; it is a different scale of event. Third, the waves that kept growing — theme share five years after the surge at or above its surge level — carry *more* excess failure (+4.3 points, 22 episodes) than the waves that receded (+1.0, 39 episodes). That runs against reading the excess as pure fad mortality, in which the collapsing waves should be the deadly ones, and against under-response, in which joiners of durable waves should thrive. A reading consistent with all three facts is selection at the gate rather than crowding out: a surging theme draws registrations for products that were never brought to use at higher rates wherever the theme is commercially active, and the theme's own subsequent growth does not rescue them. The class-and-cohort expectation these excesses are measured against, and the small number of growing episodes, both bound how far that reading should be carried.

The comparison across scales bears on what the lead penalty is. Inside a surging theme, early and late filings fail alike. The steady within-theme penalty of Section 3 — two to five points, in every era, for filings ahead of their own theme's language — is a separate fact, present where there is no surge at all. The record therefore shows two costs, not one: being in a crowded wave, which is indifferent to who arrived first, and being early relative to language that is not surging, which is not.

5 Funding, listing, and the firms that mattered

This section follows the owner rather than the mark. The unit is an owner's first registered filing, its debut, scored on the production measure, and the outcomes are dated events in the SEC record: a priced private offering (Form D), the start of financial reporting, and an initial public offering. Form D became mandatory in electronic form on 16 March 2009, so the ladder is built on debuts made 2009–2018, for which every rung is observed. The public-reporting result on the longer 1995–2018 sample, and the comparison with patenting, follow.

5.1 The ladder: a round, reporting, a listing

Of 999,442 owners whose first registered filing was made 2009–2018, 17,115 (1.71%) raised a priced private round after the debut, 4,383 (0.44%) appear in SEC reporting records at any time, and 1,795 (0.18%) registered a class of securities or completed an initial public offering at any time. The funded rung requires the round to postdate the debut; the reporting and listing rungs do not, so owners already reporting before their first trademark filing are counted, and the section's staged analysis shows they are a majority of the exchange-listed. The rungs nest: 7.3% of funded owners reach reporting against 0.3% of the unfunded, and 41% of reporting owners reach an IPO.

Table 8 gives the rate at each rung by fifth of lead and of atypicality, with fifths cut within Nice class, debut year, and fifth of description length, so that neither the industry, the year, nor the length of the description enters the contrast. Funding does not vary with either axis: the most leading fifth of debuts is funded at 1.80% against 1.86% for the most lagging, and the most atypical at 1.64% against 1.68%, both inside sampling error. Reporting and listing do vary, and in the direction Section 5.4 found on the longer sample for lead. Owners whose debut language was most lagging reach reporting at 0.59%, owners whose debut language was most leading at 0.38%; for an IPO the rates are 0.24% and 0.17%. On atypicality the sign is the other way on this sample: the most atypical fifth reports at 0.53% against 0.42% for the most conventional, and lists at 0.22% against 0.15%. Section 5.4 absorbs description length continuously on 1995–2018 debuts and finds the atypicality gradient flat; with length held in fifths on 2009–2018 debuts it is positive at t above 4. The two are stated side by side; the difference between them is a length control and a sample, and the atypicality result at the listing margin is the one claim in the paper that depends on which is used.

The lead profile at reporting and at IPO is not monotone: it falls from the lagging end through the fourth fifth and turns up in the fifth. Owners whose first description used language the class was leaving behind are the likeliest to report and to list. Most of them are established firms making a first trademark filing late in a business already built on the class's older vocabulary, and the ladder cannot separate that from anything the language itself does.

5.2 The firms that produced the most value

Among the 1.80 million owners whose first registered filing was made 1995–2018, 4,710 can be linked to SEC financial statements with reported revenue, and their peak annual revenues sum to $19.2 trillion. Ten of them hold 18.5% of that total, fifty hold 41.5%, a hundred hold 53.6%, five hundred hold 82.9%, and a thousand hold 92.6%. Whatever the trademark record says about the firms that came to matter, it says it about a few hundred owners.

Their debut language was ordinary. Standardising lead and atypicality within class and debut year, the ten largest firms' first registered descriptions sit 0.35 standard deviations below the average in atypicality and within 0.01 of it in lead; the fifty largest sit 0.08 below in atypicality and 0.13 above in lead; the five hundred largest are within 0.1 of the average on both. Weighting every linked firm by its revenue, the linked set as a whole sits 0.02 standard deviations below the average in atypicality and at the average in lead. The firms that produced most of the value described their first product in language their industry already used, and no more ahead of it than the rest.

5.3 Internet, blockchain, and artificial intelligence

Three vocabularies were hand-curated from the goods/services text: internet (the internet, the web, websites, online provision, electronic commerce), artificial intelligence (artificial intelligence, machine learning, deep learning, neural networks, natural language processing, computer vision, chatbots, generative AI, large language models), and blockchain (blockchain, cryptocurrency, crypto assets and tokens, bitcoin, non-fungible tokens, distributed ledgers, digital and virtual currency, smart contracts). Counted as class-filing rows over the full corpus, with a multi-class filing counted once per class, internet vocabulary appears in 2.94 million rows, AI in 167 thousand, blockchain in 114 thousand. Registration rates differ: 52.1% of internet filings registered, 34.5% of AI filings, 31.2% of blockchain filings, against 57.6% of all debut filings; the last two are recent and many are still in prosecution or abandoned before use. 31.6% of internet registrations carry a first-gate cancellation against 6.5% of AI registrations and 6.8% of blockchain ones; these denominators include registrations too young to have faced the gate, which is why the AI and blockchain figures are low, and they are not survival rates.

On the 2009–2018 debut ladder the three behave alike in one respect and differ in another. AI debuts are funded at 7.8% and blockchain debuts at 8.4%, against 2.9% for internet debuts and 1.7% for all debuts: a first filing in AI or blockchain vocabulary is four to five times as likely to be followed by a priced round. Reporting and listing do not scale the same way: AI debuts report at 0.66% and list at 0.38%, blockchain at 1.08% and 0.36%, internet at 0.52% and 0.22%, against 0.44% and 0.18% overall. The funding gap is a factor of five; the listing gap a factor of two, on small counts (1,824 AI debuts, 1,390 blockchain).

Patenting and funding do not order each other. Among the 8,760 owners on the patent panel who also raised a priced round, the first round came before the first patent grant for 49.9%, after it for 41.8%, and in the same year for the rest; the median lag is zero years. Neither event predicts the other's timing.

5.4 Leading debuts reach public reporting less often; unusual ones no more often

The outcome here is whether an owner ever appears in the Securities and Exchange Commission's financial reporting records. That is broader than exchange-listed: roughly half the matched firms carry a ticker and the rest report for other reasons. The rate rises monotonically in the KL *levels* on both axes, from 0.19% at the bottom past-facing quintile to 0.52% at the top. In signed ΔKL it is U-shaped.

The raw atypicality gradient does not survive the design-based estimates (Table 9). Within class and filing year, holding description length fixed, top-quintile atypicality raises the probability of SEC reporting by +0.000pp on a 0.481% base ($t = 0.0$), and by -0.003pp on the past level alone. The quintile profile is not even monotone: the three middle quintiles sit *below* the bottom one and the top merely returns to it. The 2.7-fold gradient in the raw bins is composition — firms that go on to report cluster in particular classes, years, and multi-class filings — and class×year fixed effects absorb it. Lead costs, at equal atypicality: the signed component enters negatively (-0.047pp per standard deviation, $t = -7.8$) while the unsigned component enters positively (+0.067pp per standard deviation, $t = 7.7$). The U-shape in raw bins is exactly this pair: lagging filings show the atypicality premium with no lead penalty, leading filings show it net of that penalty, and the middle quintiles show neither. Once the levels are controlled, the apparent recovery of the far-leading tail is much reduced: the quintile profile falls through Q4 and recovers only slightly at Q5 (Table 9 middle panel).

Owners are matched to SEC registrants by normalized exact name, over a candidate universe of all 5.67 million owners in the corpus; names resolving to more than one registrant are dropped rather than assigned, and one match is kept per owner string. That yields 19,889 matched owners covering 7,588 registrants, and a base rate of 0.481% among registered debut filers. Validation on 22 companies that indisputably went public — across software, pharmaceuticals, food, apparel, aerospace and autos — matches all 22. A random sample of 25 matches carrying at least three filings was inspected by hand and none was wrong, which is a check on gross failure modes rather than a precision estimate.

The linkage is nonetheless a name match, and that bounds the atypicality reading in particular. Firms with distinctive names are easier to match than firms with generic ones, and distinctiveness of a name plausibly travels with distinctiveness of the goods/services text; nothing in this design separates the two. Restricting the candidate universe — to a subset of classes, say — makes matchability itself a function of the regressor, and the atypicality gradient is sensitive to exactly that. Its absence here is therefore the more credible reading of the two, and the signed axis, which does not move when the linkage is varied, carries the weight. The financing rows of Table 10 rest on a Form D match whose raw extracts are not in the current tree and should be read as provisional.

In raw rates, 0.444% of the most lagging fifth of registered debut owners later appear in SEC reporting records, against 0.240% in the middle fifth and 0.390% of the most leading fifth: both tails beat the middle, and the lagging tail beats the leading one. Under the design-based specification the profile declines from the lagging end through the fourth quintile and recovers slightly at the leading end, so the ordering is monotone over four of the five quintiles.

The profile is a U rather than a slope, so a linear summary of it points downward through a curve whose tails are its high points; Section 6 fits it: a V with its vertex about a third of a standard deviation above mean lead, explaining 59% of the binned variation against 13% for a straight line. And "SEC reporting" is not "went public": splitting matched owners on whether their SEC filer number carries a ticker gives 10,556 listed against 9,333 reporting without one, and both halves show the same U, so the result is not an artifact of counting non-operating registrants as successes.

Atypicality predicts survival at the use-proof gate but not which owners become SEC-reporting once composition is absorbed. Lead is the timing bet, and it costs at both margins.

The two are not independent in the obvious economic sense. What makes a filing leading is precisely that its industry moved toward its language, and an industry that has moved toward your language now describes its goods the way you describe yours. Being early therefore consumes the quantity that pays: the mark that was unusual in 2011 is ordinary by 2016 not because its owner changed anything but because the category arrived. On this reading atypicality is a position competitors erode, and lead measures how fast. The design cannot demonstrate that mechanism, since it cannot separate a category catching up with a filing from some third thing moving both. What can be said is weaker: a negative coefficient on lead at equal atypicality is what the mechanism would produce if it were real.

5.5 The construct is not patenting

Within the firms that do both, trademark vocabulary and patenting move independently. On a panel of trademark owners name-matched to PatentsView patent assignees (PatentsView, 2024), each axis is regressed on $\log(1 + \text{patents})$ with firms absorbed and errors clustered on the firm, estimated on 94,181 firm-years across 17,506 firms under theme scoring (93,551 and 17,415 under term scoring) — 134,075 matched firm-years before dropping the firm-years that contribute no within-firm variation. The outcome is standardized, so β is the shift in standard deviations of the axis per unit of log-patents; one unit is roughly a tripling of the patent count, and a firm going from none to twenty moves about three units.

Text counted as	Axis	β (sd)	t	Between-firm r
Themes	atypicality	+0.012	+3.0	—
Themes	lead	−0.058	−9.4	−0.020
Terms	atypicality	−0.055	−11.5	—
Terms	lead	+0.012	+2.0	—

The grid is close to a mirror: whichever axis carries the small signal under one way of counting the text carries the opposite sign under the other. That is worth stating plainly because the two halves are easy to mistake for each other — the theme-scored *lead* coefficient (−0.058) and the term-scored *atypicality* coefficient (−0.055) are nearly the same number attached to different quantities.

Nothing here reaches a tenth of a standard deviation. Taking the largest, a firm moving from no patents to twenty shifts its mean theme-scored lead by about 0.18 standard deviations; the between-firm correlation of firm means is −0.020, and the raw firm-year correlation +0.010. The weighting of the reference window is irrelevant to all of it: scoring against an exponentially decayed reference instead of a flat one moves every coefficient by less than 0.004.

One objection is timing: a patent is applied for before its invention is in public use and a trademark when the firm is ready to sell, so patents would lead. Firms do not proceed that linearly, and in some industries the trademark comes first (Seip et al., 2018). Entering patents at $t-2$, $t-1$, t and $t+1$ — the last allowing the trademark to precede the patent — one offset at a time, since the count is strongly autocorrelated within firm, gives −0.054 to −0.089 standard deviations of log patenting under theme scoring and +0.012 to +0.024 under term scoring. No ordering isolates a direction: the offsets are flat within each scoring and opposite in sign between them. The largest coefficient anywhere is under a tenth of a standard deviation.

The second objection is heterogeneity: a pooled near-zero is consistent with genuine orthogonality everywhere, but equally with a positive slope in sectors where the two rights are complements — pharmaceuticals, chemicals, medical devices, machinery — offsetting a zero or negative slope where trademarks do the work alone. Classes are sorted in advance into complements, substitutes and neither, by judgement from the class definitions rather than by anything estimated: the eleven covering chemicals, pharmaceuticals, machinery, electronics, medical apparatus and scientific services are complements, the consumer-goods and services classes substitutes, the rest neither. The sectoral build differs from the pooled grid: it is scored at 200 themes, keeps firms with at least three panel years, and requires 50 firms and 300 firm-years per cell. Estimating inside each group gives -0.000 ($t = -0.03$) among complements, -0.026 ($t = -1.45$) among substitutes, and an interaction of -0.008 ($t = -0.49$). Across the 27 Nice classes clearing a 50-owner floor, the complement classes run from sixth place to twenty-fourth by coefficient and do not agree in sign with one another. Within that 27-class set the only cells that reach significance are paper and housewares, both negative and neither a complement class. On the wider 34-class retention used for the sectoral build, the significant positives are transport and cosmetics — again neither a complement — while medical devices and machinery turn up among the significant negatives. Neither retention supports the complementarity story.

The test is conditional on a successful assignee match, so every owner patents at least once and the complement and substitute cells resemble each other more than the industries they stand for do, pushing any interaction toward zero. The weight rests on the cell-level nulls — among the firms where a patent–vocabulary relationship should be easiest to find.

5.6 Each axis bites at a different stage

Table 10 re-estimates every outcome on one harmonized sample — one row per debut owner, scored on that owner's first filing — under a single specification, so the coefficients can be read down a column. Magnitudes differ from the section-specific tables above.

Surprising language costs at registration and pays afterwards: atypical debut filings complete registration less often (−1.35pp per standard deviation), then survive the first use-proof gate more often (+1.35pp) and reach public markets more often (+0.04pp on a 0.56% base — the one specification in which a positive atypicality coefficient on listing survives, and it is a continuous term through a non-monotone profile, not the quintile contrast of Table 9). Surprising filings register less often; among granted marks the market does not. Having been early runs the other way: leading applications complete registration *more* often (+0.87pp) and then pass the use-proof gate *less* often (−0.49pp on passing). Both survival rows point the same way at the year-ten renewal, four years later and on a sixth of the sample: atypicality +1.30pp and lead −0.63pp among marks that had already cleared the first gate. Whatever the first gate is selecting on, the second gate selects on it again rather than reversing it. Those filers follow through on the paperwork; what does not follow through is the product.

A linear coefficient cannot represent the inverse-U documented in Section 3.1, where both vocabulary tails complete registration less often than the middle. The positive linear term reflects the asymmetry of that shape — the leading tail completes more often than the lagging tail — and not a monotone gain from leading.

In the financing rows atypicality does nothing at one stage and a great deal at the next. Whether a debut owner ever raises a Regulation D round — a private securities placement, reported to the SEC on Form D — is unrelated to how atypical its filing was: −0.017pp per standard deviation among owners who completed registration, against a 2.61% base, which is indistinguishable from zero ($t = -0.8$). Among owners who did raise one, the same measure predicts *reaching* the public market: +0.49pp on a 3.49% base, about 14% relative ($t = 3.8$). Atypical language is invisible at the financing stage and informative afterwards, which is what an investor screen that does not read the text would look like — the text being free and public at the moment of filing. The comparison is among survivors, so it cannot separate that reading from selection on unobservables; and the null at entry is a null, not evidence of indifference.[8]

[8] This pair is sensitive to the reference window. Under annual reference buckets the entry row was −0.06pp ($t = -2.8$), which read as active negative selection. Per-filing windows move it to zero and leave the listing row intact, so the negative selection was an artifact and the informativeness among the funded was not.

That last row turns on one construction. An exchange-registration marker on its own does not mean a firm listed *after* raising: 56% of matched owners carrying one registered before their first Regulation D notice, because established public companies also place securities privately. The row is therefore restricted to owners whose exchange registration postdates their first notice, which halves the coefficient and leaves it significant.

Three limits bound the financing rows. Regulation D notices are electronically filed only from 2009, so they rest on 2009–2018 debut cohorts. Owners are matched by normalized name, which succeeds for a small and unrepresentative minority favouring formally constituted entities, so every estimate is conditional on matchability. And the final row conditions on funding, realized after the filing being scored, so the coefficient is a contrast among survivors rather than a treatment effect: it establishes that information distinguishing eventual listers was already present in the text years earlier, not that the language caused the outcome.

Lead, by contrast, does not reverse anywhere after registration. It is negative at both use-proof gates, at financing with or without registration imposed, and at listing among the funded; the only row where it is not negative is SEC reporting among registered owners, where it is indistinguishable from zero (-0.004pp, SE 0.007). Leading filings fare worse at every stage where the market, rather than the examiner, is doing the selecting, and never rewarded at any of them.

6 Robustness: what changes under other choices

Each result above is stated under one formulation chosen for legibility: one theme resolution, flat weighting inside the reference windows, symmetric five-year windows, estimates conditional on grant, and the 2002–2018 registration cohorts. This section reports what each of those choices costs, and in each case states the more complicated finding beside the simpler one the body uses.

6.1 Representation, anchoring, weighting

A scoring is fixed by two choices. The *representation* is what the language is turned into: a distribution over 50 fitted themes, or a distribution over the class vocabulary's terms. The *reference* is what that is compared against, and it hides two sub-choices — the *anchoring* (one reference per class-year, or one per filing built on its own date) and the *weighting* inside the window (every day equal, or decaying by half every two years). Six builds exist:

Representation	Anchoring	Weighting	sd(lead)	sd(atypicality)
Themes	class-year	flat	0.132	0.672
Themes	per-filing	flat	0.102	0.698
Themes	per-filing	half-life 2y	0.082	0.698
Terms	class-year	half-life 2y	0.299	1.330
Terms	per-filing	flat	0.155	1.101
Terms	per-filing	half-life 2y	0.130	1.102

All six are compared on the 6.01 million filings every one of them scored. The second row is the production scoring. The fourth is the legacy build, and it is the one to handle carefully: it moves anchoring *and* weighting at once, so a result computed on it cannot be attributed to either. That is why the two per-filing term builds were constructed.

Changing one choice at a time settles the ordering. Correlations of lead between builds differing in exactly one respect:

Choice varied	r (lead)	r (atypicality)
Weighting, flat vs half-life 2y (themes)	0.982	1.000
Weighting, flat vs half-life 2y (terms)	0.970	1.000
Anchoring, class-year vs per-filing (themes)	0.738	0.929
Resolution, 50 vs 200 themes	0.636	0.828
Representation, themes vs terms	0.460	0.396

The weighting is close to irrelevant: whether recent language counts for more than distant language within the window barely moves the score, and it does not move atypicality at all to three decimals, which is what the 45-degree rotation of Section 2 implies — the long axis is a property of the filing and the weighting acts almost entirely on the small residual. Anchoring matters moderately. Resolution and representation matter most: quadrupling the number of themes leaves the two builds agreeing on 40% of the variance in lead, and themes against terms on less than a quarter. A robustness check that varies only the weighting is therefore close to a restatement; one that varies the representation or the resolution is a real test, which is why the term-scored replication in Appendix 6.11 and the sweep below are the ones the claims lean on.

6.2 Theme resolution

Fifty themes over a 62,168-term vocabulary is roughly one theme per industry, so the partition is coarse enough that a reader may reasonably suspect the findings belong to it. Raising the theme count does not add coverage — the vocabulary is fixed, and a larger T re-partitions the same words — but it changes what counts as sitting far from an industry, and could change any sign. Re-fitting the model at four resolutions on the same stratified sample and the same analyzer, and re-scoring the software class under per-filing windows:

Themes	Gate contrast, lead	t	Gate contrast, atypicality	r with $T = 50$
50	+3.23pp	13.8	−14.01pp	—
100	+2.71pp	11.6	−13.69pp	0.690
200	+5.20pp	22.2	−13.45pp	0.640
500	+5.69pp	24.3	−14.49pp	0.641

Three readings, in descending order of how much weight they carry. The sign and the significance are untouched across a tenfold change in resolution, and so is atypicality, which stays within a point of −14pp throughout. The magnitude is not: the lead contrast runs from +2.71 to +5.69pp and does not move monotonically in T, so no single resolution can be quoted as the software class's effect, and the figure this paper reports at $T = 50$ sits at the low end of that range rather than the high one. And agreement with $T = 50$ settles at 0.64 without decaying as T grows, which says the resolutions share a fixed core and differ in the remainder rather than drifting apart as the partition refines.

The corpus-wide contrast is steadier than the single class: at $T = 200$, scored for all 45 classes, it is +2.23pp against +2.29pp at $T = 50$. Only those two resolutions exist corpus-wide, so that is a two-point comparison, but it is the one the paper's estimates run on, and pooling evidently averages away much of the per-class resolution sensitivity.

6.3 Fitting noise at one resolution

The sweep leaves one question open. Scores at different resolutions agree at $r = 0.64$ to 0.69 on lead, and neighbouring resolutions agree no better than distant ones. That has two readings. The number of themes may genuinely change what is being measured. Or a theme-model fit may simply be unstable — two runs settle in different local optima — in which case the sweep is measuring fitting noise and the resolution has little to do with it.

Holding T fixed and moving only the seed separates the two. A second fit at $T = 200$ on the same sample, the same vocabulary and the same number of passes, differing from the first in nothing but its random seed, agrees with it at $r = 0.79$ on lead and $r = 0.87$ on atypicality across 1.11 million scored software-class filings. The cross-resolution figures were 0.64 and 0.83. Most of what the sweep showed as resolution sensitivity is therefore present between two fits that differ only in the seed: for atypicality nearly all of it, for lead well over half. The resolutions are not measuring different things; each fit is a noisy draw on the same thing.

The estimate moves less than the scores do. On the identical gate sample, the first-gate lead contrast is +5.20pp under the first seed and +4.88pp under the second (SE 0.23 each), and the atypicality contrast −13.45 and −14.38pp. Two consequences follow. A single fit's lead score carries measurement error of roughly a fifth of its variance, which attenuates every coefficient on lead toward zero; the estimates in this paper are on the conservative side of what a noiseless score would return. And the spread of magnitudes across the resolution table above, which does not move monotonically in T, should be read as a band about a third of a point wide from fitting noise alone rather than as a property of any particular T.

Annual anchoring leaves a signature that per-filing anchoring does not. Mean lead by calendar month of filing, in units of the build's own standard deviation, has no reason to trend — nothing about filing in December rather than January makes a description more forward-looking:

Representation	Anchoring	December − January (sd)
Terms	class-year	+0.169
Themes	class-year	+0.041
Themes	per-filing	−0.008
Terms	per-filing	+0.017

A December filing sits eleven months further from its past reference and eleven months nearer its future one, and under annual buckets that shows up as position in the calendar. Under per-filing references it is gone.

The substantive claim does not depend on any of this. The first-gate contrast, computed on the common sample, runs +2.73, +2.71, +2.39, +4.05, +3.97 and +3.19pp on lead across the six builds in the order tabulated above, and −7.75 to −6.17pp on atypicality. Every sign holds; term scoring returns the larger lead contrast, so the headline estimate is the conservative representation.

6.4 Decayed reference windows

Inside the reference windows every day counts the same. The alternative weights recent language up, with a half-life of two years inside the same 1,826-day truncation, so that a term the class adopted last month is more familiar than one it adopted four years ago. Rescoring the corpus that way changes the scores little and the estimate somewhat. Flat and decayed lead correlate at 0.980 on the 3.10 million gate registrations and atypicality at 1.000 to three decimals. On the identical registrations, with quintiles cut within class and registration year, the top-minus-bottom gate contrast on lead is +2.29pp under the flat window and +1.95pp under the decayed one (SE 0.09 each); on atypicality it is −4.39 and −4.43pp. The decayed window therefore returns a lead penalty about a seventh smaller, with the same sign, the same precision, and the same profile. The flat window is kept in the body because it is the simpler statement; a reader who prefers recent language to count for more should take +1.95pp as the estimate.

6.5 Annual reference buckets

Every estimate uses a reference anchored on the filing's own date. The alternative is one reference per class-year, under which a filing made on 1 January and one made on 31 December receive identical references and neither sees any of the language filed during its own year. Appendix 6 reports the calendar signature that leaves.

First, the gate penalty was overstated by about a third. The persisted comparison is the software class on an identical 452,911-registration sample: +7.40pp (SE 0.43) under annual buckets against +5.25pp (SE 0.42) under per-filing windows, a ratio of 0.71, with the t-statistic *rising* because the standard error falls further than the estimate. The corpus-wide annual regression of the superseded build returned +2.15pp (SE 0.30) against the current +1.39pp, the same ratio, and its per-split ratios ran 0.59 to 0.73; that run was not persisted, so the corpus-wide statement rests on the superseded build's log and the software-class artifact carries the claim.

Second, the shape of the gate penalty changes. Under annual buckets the quintile profile looked like a gradient with a steeper final step; under per-filing windows the gradient through the interior quintiles is shallower and the final step dominates, with the top quintile carrying about seventy percent of the total. The coarser measure spread a top-quintile effect across the whole range.

Third, one published shape does not survive. Registration completion showed an inverse-U on both the unsigned and the signed axis. Only the unsigned one is real; on ΔKL the curvature disappears entirely under per-filing windows, and the earlier interior peak is attributable to the month gradient above. The firm-level results are the least affected by the window change, though they are sensitive to the owner-to-registrant linkage instead (Section 5.4): the signed decomposition keeps both its signs and its significance under either window.

6.6 Window width

The window is a parameter of the measure, not a property of it, and the five years used throughout was inherited from the class-year construction rather than chosen under the current one. Re-scoring the software class at several widths under per-filing references, and judging each by the first-gate quintile contrast it produces, gives +2.02pp at two years, +2.20 at three, +3.23 at five and +3.37 at seven, each on an identical sample of 452,911 registrations and each with a standard error of 0.23pp. Signal rises steeply from two to five years and is flat between five and seven: the +1.03pp gain from three to five is well outside the sampling error, the +0.14pp from five to seven is not. A ten-year window returns +2.89pp, but that figure is not comparable to the others — requiring both windows to lie inside the corpus costs 22% of the scored filings and 13% of the gate sample, dropping the earliest and latest registration years, which are also the years whose contrasts differ most. The apparent decline at ten cannot be separated from that change in composition and is not evidence about window width. On the range where the sample is fixed, seven years is nominally the peak and five is within noise of it while retaining more of the corpus, so five is defensible rather than optimal. The sweep is on one class and one outcome.

6.7 Mixing the two windows

"Leading" is relative to a reference, and the two references need not be the same width. A short past and a long future make lead a measure of foresight — whether the filing resembles where the class was heading over the next seven years more than what it wrote in the last three. A long past and a short future make it a measure of rupture — whether the filing has broken with what the class had established over seven years, judged against only the next three. Scoring every class on the production representation and per-filing windows at $W \in \{3, 5, 7\}$ years on each side, and evaluating all five variants on the identical 3.10 million registrations with quintiles assigned within class×registration-year:

Variant	Past, future (years)	First-gate failure Q5−Q1 (pp)	First-gate failure classes > 0	Registration Q5−Q1 (pp)	Registration Q3 − tails
Foresight	3, 7	+1.65 (0.09)	31 of 45	+1.09	+0.50
Symmetric	3, 3	+1.69 (0.09)	—	+0.79	+0.13
Symmetric	5, 5	+2.29 (0.09)	33 of 45	+0.71	+0.28
Symmetric	7, 7	+2.58 (0.09)	—	+0.48	+0.65
Past-rupture	7, 3	+3.00 (0.09)	37 of 45	+0.01	+0.33

The symmetric five-year row is the paper's FE-only estimate in Table 3, reproduced here to the decimal, which fixes the scale. The penalty rises monotonically along the mix, from +1.65pp under foresight to +3.00pp under rupture, and rupture is the more pervasive variant, positive in 37 of 44 classes against 31. In software the two are +1.82 and +5.07pp. So the use requirement punishes distance from the class's established language more than it punishes resemblance to its coming language; but foresight carries a penalty of its own, more than half the symmetric one, and the separation is a matter of degree rather than kind.

Registration is a different matter. On the signed axis the top-to-bottom contrast is at most a point on a 57% base and the interior-minus-tails depth at most two thirds of a point, under every mix. An earlier version of this decomposition, run on the retired term-level calendar-year scoring, found a registration inverse-U of four to five points that the window mix did not touch; under per-filing references that curvature is gone on the signed axis, consistent with Appendix 6.5, which traces it to the calendar artefact. The examiner's response to lead, on this scoring, is close to nil.

6.8 Without conditioning on grant

Figure 9 repeats the lifecycle outcome analysis without the registration conditioning, on filings 2013–2015 pooled across all classes (a window whose registrations land by ∼2017 and face at least one Section 8 gate by the 2025 snapshot; the earliest registrations in the window also face their year-ten renewal, so panel B mixes one and two gates). Panel A is the unconditional composite, the probability that a filing both registers and passes the gate, measured over all filings including the never-registered. It is nearly flat in ΔKL — 22.4% at the bottom quintile against 22.1% at the top, a middle-versus-tails depth of +0.001pp (SE 0.10) — because two opposed gradients cancel: registration rises in lead (61.3% to 63.2%) while gate survival falls (44.2% to 40.8%). Conditioning on grant is what separates them. Panel B shows the conditional gate outcome on the same filings, and it is U-shaped rather than inverse-U (depth −2.11pp), with a clearly penalised leading tail consistent with the cleaner single-gate cohort of Figure 10.

6.9 Changing the time period

The registration cohorts the gate estimates use begin in 2002 because that is the conventional cut for this corpus, and end in 2018 because the cancellation record runs to April 2026. Section 3.5 carries the series back to the first scoreable cohorts: the penalty was two to three times larger for marks filed in the late 1990s than for any cohort since, reversed for filings made 2000–2004, and returned from 2008, with the swing inside the four technology classes and, within them, in which themes each era's leading filings were drawn from. Within the 2002–2018 window the penalty is zero to negative for the 2002–2005 cohorts (−2.3pp for 2003) and runs +1.8 to +5.4pp for every cohort from 2008 on the wide-window raw series. The 2019 cohort, evaluable only on the narrow

6.5–7.0 observation band for registrations issued early enough in that year (78,408), replicates at +5.6pp. Cohorts from 2020 have no observed gate. The pooled estimate is therefore an average over a penalty that was not constant, and a reader who wants the penalty for a period should read Figure 4 rather than the pooled row.

6.10 Lead is a small residual

Past-facing and future-facing KL are highly correlated however scored. On the complete class-009 corpus under the production scoring they correlate at $r = +0.988$ (n = 1,109,643); under term scoring on class-year references the figure is +0.971. The lead L is therefore a small residual of two large, stable quantities: both levels have a standard deviation near 0.70 nats about a mean of 1.46, while L has a standard deviation of 0.109 — about a sixth of either level's spread. Only 1.2% of the two levels' combined variance survives into their difference. That is what makes L sensitive to anything that inflates the levels, and under term scoring the dominant such thing is document length. Term-scored atypicality correlates with the log of a filing's distinct term count at −0.681 in class 009 and −0.620 in class 035, for the reason set out in Section 2: the divergence charges a filing for the narrowness of its own distribution, and that narrowness is fixed by how many distinct terms it uses. Theme scoring weakens the coupling, taking the same correlations to −0.217 and −0.098. The consequence for the lead is direct: the standard deviation of L rises 2.2–2.5-fold across deciles of A under term scoring and 1.5–1.8-fold under theme scoring, so a unit of lead means roughly the same thing everywhere only in the theme representation. Appendix D shows this as a length surface over the two axes. Two qualifications. Theme scoring returns a finite value for about 65% of filings, so the two columns are not drawn from the same set; re-estimating the term-scored correlations on exactly the theme-scored subset barely moves them, which puts the contrast down to the representation rather than the sample. And "flat" means no monotone length gradient rather than constant: class 035 under theme scoring is mildly U-shaped in A. The two scorings agree only moderately per filing (Pearson 0.46, Spearman 0.42 on the 6.01 million filings both score), which gives their agreement on the mark-level patterns evidential force and their disagreement on the firm-level patterns diagnostic force (Appendix 6.12). Across $W \in \{3, 5, 7\}$, 6.9–10.4% of filings flip sign. Per-filing noise attenuates at every aggregation used: firm-year means, token pools, theme prevalence trajectories.

Alignment with expert innovation judgment is at most suggestive. Firms on BCG/MIT "most innovative" lists sit +0.09 standard deviations above the panel mean under term scoring ($p = 0.22$), +0.14 under theme scoring ($p = 0.07$), and +0.08 at $T = 200$ ($p = 0.20$). The matched sample is 41 firms, which is too small to carry weight in either direction, and the paper draws no inference from it. Patent counts remain uncorrelated with ΔKL on the same panel (Pearson +0.010).

6.11 Term scoring as a cross-check

The snapshot cross-check on the single one-gate-elapsed cohort (registrations 2016–2018, terminal status 710 versus 701–705/800) agrees with the event-dated estimates and is scoring-robust: −5.4pp survival across quintiles under theme scoring and −5.7pp under term scoring on the identical sample (Figure 10). Under term scoring, gate survival *rises* in the raw KL level (+6.5pp past-facing, +7.6pp future-facing): atypical text survives the use requirement better while leading text survives it worse — the same two-axis separation as Section 5.4, at the mark level.

6.12 Term scoring at the firm level

Under term-level scoring, leading filings appear to belong to better firms. Among registered debut filings, the probability that the owner ever appears in SEC EDGAR rises in term-ΔKL from 0.29% at the bottom quintile to 0.42% at the top; on an SEC-matched firm-year panel, trailing 3-year mean term-ΔKL predicts gross margin at +2.3pp per standard deviation ($t = 2.7$ on the locally rebuilt panel through 2019; +3.2pp, $t = 5.3$, on the published panel through 2024), with past-facing surprise entering positively and future-facing negatively; and firm-year term-ΔKL predicts excess stock returns of +1.5 to +5.0pp per standard deviation at one-to-four-year horizons. A larger debut-only return figure (+28pp at 4 years) fails a sample-size diagnostic and is not claimed.

None of the signed versions survives distribution scoring. On identical samples, the debut listing gradient in signed theme-ΔKL is flat to declining, and the margin coefficient reverses: −1.3pp per standard deviation ($t = -1.8$) at $T = 50$, −2.2pp ($t = -3.1$) at $T = 200$, and −0.8pp ($t = -1.1$) at $T = 500$, against the term-level +2.3pp; the past-/future-facing decomposition flips sign and stays flipped. The term-level listing gradient separately dissolves under document-length controls: within length bands it is U-shaped rather than rising, with the monotone rise contributed by composition across lengths (listed firms' counsel write longer descriptions). The mark-level results are unaffected — they hold under both scorings at every resolution.

Vocabulary blindness does not rescue the signed signal. The theme model's vocabulary is sample-built, so rare and emergent terms are invisible to it; if the firm signal lived exactly there, theme scoring would erase a real signal rather than an artifact. Computing each filing's invisible-vocabulary share (63.8% of class-009 term *types* but only ∼9% of token *mass*): the share itself is a strong firm marker (P(EDGAR) 2.9% to 5.5% across its quintiles; gate survival +5.7pp), but *within* invisible-share strata the signed ΔKL gradient on the firm outcome is flat to negative, while the gate penalty holds in both strata (−6.9 and −8.3pp). The term-level firm signal is *unsigned specificity*: serious firms describe specific products with specific, therefore rare, words, and rare-term mass inflates term-ΔKL mechanically. Theme count is a partition dial, not a coverage dial — a crypto/blockchain theme exists at $T = 200$ but not at $T = 50$ or $T = 500$, because raising T re-partitions the fixed vocabulary along dominant axes — so resolution sweeps test granularity, not coverage, and the invisible-share split is the test that binds.

The registration filter does not explain the divergence. Dropping the registration conditioning leaves the term-level listing gradient essentially unchanged (0.26% to 0.38% across quintiles against 0.29% to 0.42% conditional), and an examiner channel touching 4.3% of failed applications cannot generate several-point quintile gaps.

Term-resolved text-novelty measures can therefore manufacture firm-performance correlations from drafting style and lexical specificity, and a distribution-based rescoring is a cheap, decisive diagnostic.

6.13 Functional form

Four shapes were fitted to each profile by weighted least squares on 40 equal-count bins (Figure 11): linear, quadratic, an exponential $a + be^{cx}$, and a free-vertex symmetric form $a + be^{c|x-x_0|}$ meant to catch a U directly. Selection is by AIC.

Outcome	Axis	R^2 line	R^2 curve	ΔAIC	Shape selected
Registration	atypicality	0.38	0.92	77.1	curve, peak at +1.04
Registration	lead	0.08	0.17	1.9	(quadratic; weak)
Use-proof	atypicality	0.74	0.74	0.0	linear
Use-proof	lead	0.44	0.55	4.3	V, vertex −1.50
SEC reporting	atypicality	0.74	0.84	15.2	curve, peak at +2.17
SEC reporting	lead	0.13	0.59	26.3	V, vertex +0.30
listed only	lead	0.19	0.46	12.2	V, vertex +0.41
reporting, not listed	lead	0.07	0.67	37.2	V, vertex +0.24

The winning form is a free-vertex $a + b\,e^{c|x-x_0|}$. On the atypicality rows it is a genuine curve, with fitted exponents near 0.5 to 0.9 — a hump that peaks about one standard deviation above mean atypicality at registration and about two above it at SEC reporting. On every lead row the exponent collapses to the order of 10^{-4}, which turns $e^{c|x-x_0|}$ into $1 + c|x - x_0|$: what wins there is a V, linear in distance from a vertex. A V beating a parabola says the turn is sharper than a quadratic allows, not that anything is multiplying. The script now flags that degeneracy rather than leaving it to be read off the fitted constants.

Vertices are in standard deviations of the axis across filings. An earlier version of this table standardized on the spread of the forty bin centres instead, which is narrower than the filings' own spread and put the vertices on a scale that did not match the per-standard-deviation slopes quoted elsewhere in the paper.

The two rows for SEC reporting are the outcome Section 5.4 and Table 9 actually run on; the listed and reporting-only splits below them show the same shape on each half, which is why the split does not change the reading.

AIC here is computed on 40 binned means, not on the millions of underlying filings, so it ranks shapes rather than testing them. The 77 at registration and the 12 to 36 at the public-reporting gates are decisive; the 1.9 at registration on lead and the 4.3 at the use-proof gate are weak, and those two rows are better read as "no worse than linear" than as curvature.

6.14 Listing against reporting

"Appears in SEC records" mixes two populations: companies whose shares trade, and entities that file with the Commission without ever listing — private issuers, funds, subsidiaries registering securities. Splitting the 19,889 matched owner strings on whether their CIK carries a ticker in the Commission's own company-ticker file gives 10,556 listed against 9,333 that file financial statements without one.

It does not change the finding. Among registered debut owners whose debut filing is scored, 0.310% later appear as listed companies and 0.285% as reporting non-listed ones, and both profiles are U-shaped in lead with vertices at +0.41 and +0.24 standard deviations. The listed profile is slightly flatter and noisier, which is what the smaller and more selected outcome predicts. Whatever the signed axis is picking up at this margin, it is not an artifact of counting non-operating registrants as commercial successes, and it is not specific to the act of listing either.[9]

[9] Two limits on the split. The ticker file is a current snapshot, so a company that listed in 2003 and was acquired in 2011 appears in the non-listed tier; the tiers are therefore "currently listed" rather than "ever listed," which will misclassify older successes downward. And the base rates here are computed on the appendix's own panel, which conditions on the debut filing carrying a score and so is smaller than the body's sample.

6.15 Filings are not independent draws

Some of them share an exact position with another filing, and outcomes cluster within owner — the within-owner correlation is 0.38 for gate survival pooled and is mechanically 1.0 for public listing, since listing is a property of the owner and a firm with hundreds of marks contributes hundreds of identical successes. The regression tables are unaffected, because the gate regressions cluster on normalized owner and the firm-level specifications carry one observation per owner. The raw quintile contrasts of Sections 3–4 — the era, theme, internet and surge tables — are not: their two-sample standard errors ignore the owner clustering, and their t-statistics overstate precision by roughly a factor of two, which the text of those sections now carries where the margins are close. Binning filings without the correction likewise overstates how much structure the corpus contains: on a positional grid, correcting cell errors for the clustering design effect — the factor by which within-owner correlation inflates the variance of a cell mean — cuts apparent excess dispersion by roughly a quarter on the gate outcome and by half on listing. Displays of the vocabulary plane are descriptive here for that reason, and the near-zero pooled correlation between A and L reported in Section 2 does not survive cell by cell within class, so per-cell contrasts should not be read as within-class effects.

6.16 What remains unresolved

What drives the episodic pattern is not settled. Cohort and gate year move together, so post-crisis bet composition, app-era speculative filings, and calendar-time enforcement changes cannot be separated on this design. The representation is also fit on the pooled corpus; a vintage refit would close the remaining look-ahead channel.

A Corpus, scoring, and burn-in

A.1 Corpus and scoring

The USPTO Trademark Annual Applications backfile (product TRTYRAP) provides a complete record of US trademark filings from 1884 onward. The deepest diagnostic work uses the complete software/electronics class (Nice 009; 1.81M filings); the diffusion application uses four classes totalling 2.32 million granted filings 1990–2024; the gate analyses use all 45 classes. The firm-level panels link trademark owners to SEC EDGAR financial filers and to PatentsView patent assignees by normalized-name join.

The theme representation is a $T = 50$ LDA fit on a stratified sample of 448,437 filings across all 45 classes (vocabulary of 62,168 unigrams and bigrams); every filing 1990–2024 is transformed to a dense 50-theme distribution, and past-facing and future-facing surprise are computed against per-filing theme references spanning 1,826 days on each side of the filing date, scoring 1995–2019. A filing whose window holds fewer than 500 same-class filings on either side is left unscored, since a thin reference inflates the divergence mechanically; that floor, together with the corpus edges, is why roughly two-thirds of filings carry a score. A term-level variant is retained alongside, on the same anchoring: token distributions (within-class document frequency ≥ 50) against per-filing windows of the same 1,826 days, Dirichlet-smoothed ($\alpha = 1$) over the class vocabulary. Because a goods/services description runs to a few dozen words, the divergence is needed only at the terms a filing actually uses, so the reference is carried as a running count vector and read sparsely. The term variant is used where vocabulary identity is the object (phrase transit) and as the comparison scoring throughout. Both representations are also computed under an exponentially decayed reference with a two-year half-life, truncated at the same window; Appendix 6.11 reports what changes.

A.2 Scores are interpretable from 1995

Burn-in is the span at each end of the corpus where the reference windows cannot be filled, so scores there reflect a thin reference rather than the text being scored.

The signal depends on having enough past and future filings to populate the references; the 5-year past window is incomplete before 1990 and the future window after 2020, and both zones are excluded (Figure 12). The burn-in cut is set empirically: thin past references inflate past-facing surprise, and profiling that bias across all 44 sufficient-volume classes gives a median absolute bias of ∼2.0 nats — the natural-log unit in which these divergences are measured — in 1990, 0.25 in 1991, 0.10 in 1992, and 0.04 from 1993 onward, below any interpreted effect size. The 1995–1997 deviations (∼0.15–0.17) are not burn-in — mechanical contamination cannot rebound — but the dot-com era moving the corpus. Per-class burn-in lengths estimated under a stability-band criterion track each class's noise floor rather than its reference density, so a single standard cut at 1995 is used, conservative by about two years.

B Examination rewards a middling specificity

Across 3.59 million debut filings (57.6% base rate), completion is an inverse-U in *atypicality*: it climbs from 46.6% at the least atypical end to about 58% in the middle and falls back to 53.4% at the most atypical (Figure 2, panel C). The same shape appears on the past-facing level alone, more mildly, running 54.0% at the bottom quintile to 58.4–59.1% in the middle and 57.7% at the top (Figure 1A).

The signed axis behaves differently. Completion on ΔKL falls to a trough of 51.9% just above the median and then rises steadily to 58.0% at the leading end. That trough is composition rather than timing: $|L|$ correlates with atypicality at +0.31, so filings near zero lead are disproportionately low-atypicality filings, which complete poorly for the reason above. Split by atypicality, the trough is absent in the upper three fifths and survives in the lowest. Reference windows built by calendar year put an inverse-U on this axis too, but that one is an artifact of annual bucketing (Appendix 6.5) and does not survive at daily resolution. The signed axis still enters a linear specification strongly (Table 10), but monotonically rather than through an interior peak: what curvature exists at registration belongs to how far a filing's language sits from its category, not to which side of it the filing sat.

The obvious objection is that unfamiliar language is simply refused more often, so the measure recovers nothing beyond familiarity. Completion falls at *both* ends, so the most category-typical filings also complete less than the middle, which familiarity cannot explain; and the curvature is specific to the unsigned axis, the signed axis showing no interior peak at all.

Refiling after abandonment is itself mildly related to vocabulary position, though not on the axis that matters here. Refiling rates are flat in atypicality (a middle-minus-tails difference of −0.06pp on a 3.6% base) but rise with lead, from 2.6% among the most lagging abandoned applications to 3.4% among the most leading. Firms whose language ran ahead of their industry are slightly likelier to try the same mark again.

C How refiled descriptions are rewritten

Figure 13 shows the goods/services description before and after a rewrite, for the 31,084 refilings in Section 3 whose text changed. A pair is an abandoned application and the same owner's later registration of the identical mark in the same Nice class within six years; the earliest registered refiling is kept. Counsel of record is read from the prosecution history of each filing separately, so a pair can change representation between attempts. Atypicality is on the production scoring; length is the word count of the normalized description.

The means understate the convergence, because it runs in both directions. By fifth of the original's atypicality, the change is +0.16, +0.14, +0.06, −0.06 and −0.39 nats from the most conventional fifth to the most atypical, and within a point of that profile in each of the four representation groups. Identical-text refilings change by +0.002 to +0.008 across the same fifths. On lead, the changed-text profile runs +0.046 to −0.072 and the identical-text profile +0.014 to −0.054; the difference is the rewriting, the rest is the reference windows moving with the later filing date.

D Term scoring measures length; theme scoring does not

Figure 14 puts the two representations side by side over the same axes, with each hexagon coloured by the mean length of the filings in it. Under term scoring the colour runs from dark near the origin — a median of about 200 terms — to near-white at the far end, where the median is eight. That gradient is the measurement problem: a filing scores as atypical largely because it is short. Under theme scoring the same surface is a narrow band of roughly uniform tone.

Two filings are marked in both columns. Under the production scoring they behave differently from one another, which is the point. AMAZON.COM (1997) is a leader on both representations — the 98th percentile of its class on terms and the 94th on themes — while sitting at the 40th and 47th percentiles of atypicality, so both agree that it was early and neither mistakes it for strange. Apple's IPOD (2001) divides them: the 31st percentile of lead on terms against the 62nd on themes. The surfaces plotted here are on the class-year build the figure was generated from, where the length gradient is starkest; the percentiles just quoted are on the per-filing build used for every estimate.

Theme scoring is independent of document length, which term scoring is not; term scoring resolves novel word *combinations*, which theme scoring does not. Hence theme scoring for every estimate, since the length confound contaminates firm-level comparisons, and term scoring for every worked example, since the theme basis has no coordinate for the language the examples turn on.

E How three filings encode

Table 11 traces three filings through term scoring and theme scoring at $T \in \{50, 200, 500\}$ — three kinds of novelty, and how each representation handles them.

AMAZON.COM (1997, advertising/retail) is *combinatorial* novelty: every word is ordinary, but the bigrams assemble an invention ("computerized line," "line search," "search ordering"; class document frequencies 379, 134, 355). The filing's text reads "on line" as two words, the standard spelling in 1997, before *online* lexicalized into a single token; the analyzer drops the stopword "on" and the surviving bigram "line search" is the act of forming the compound caught in progress. The novelty was necessarily combinatorial because the word for it did not yet exist. Term scoring sees these bigrams; under theme scoring at every resolution all of them are out of vocabulary and the filing reads as a media retailer that uses computers.

KOMBUCHA (1997, beer/soft drinks) is *lexical* novelty: the new thing arrives as new words (*kombucha* df 567, *saccharose* out of vocabulary even at the class level). The word *kombucha* is in the theme vocabulary but, with no kombucha-like theme, its mass dissolves into a generic beverages theme (≥ 0.60 weight at every T).

Ethereum (2018, software) is *wave* novelty: emergent vocabulary already diffusing (*blockchains*, *distributed computing*). It loads on generic software and services themes at every resolution, including $T = 200$, which does contain a cryptocurrency/blockchain theme — the eleven occurrences of *software* outvote the blockchain terms.

The pattern across the row is the measurement point in miniature: theme scoring cannot represent combinatorial novelty (Amazon), dilutes lexical novelty (Kombucha), and majority-votes away wave novelty (Ethereum) even when a fitting theme exists. That the mark-level outcome results are nearly identical under both scorings despite this blindness is what licenses theme scoring as a robustness instrument; that the firm-level results diverge is what exposes them as artifacts of the lexical specificity term scoring additionally captures (Appendix 6.12).

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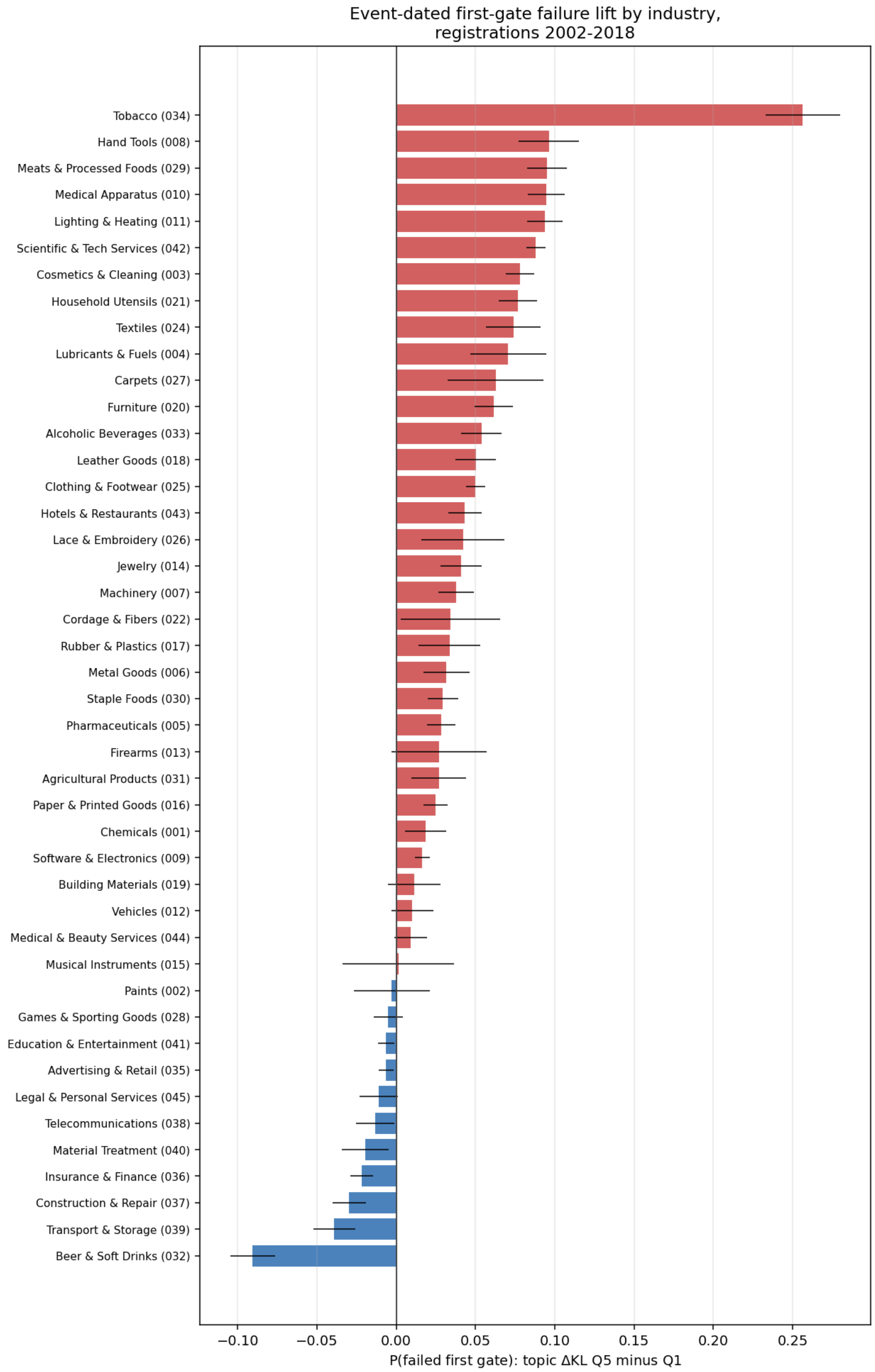

Figure 3: Event-dated first-gate failure lift (theme ΔKL top minus bottom quintile) by industry, registrations 2002–2018, raw quintile contrasts. Positive bars indicate the leading penalty. Whiskers are two-sample binomial intervals, uncorrected for the within-owner correlation (0.38) and so narrower than clustered intervals; read Table 3 for precision. Per-class breakouts are in the online appendix.

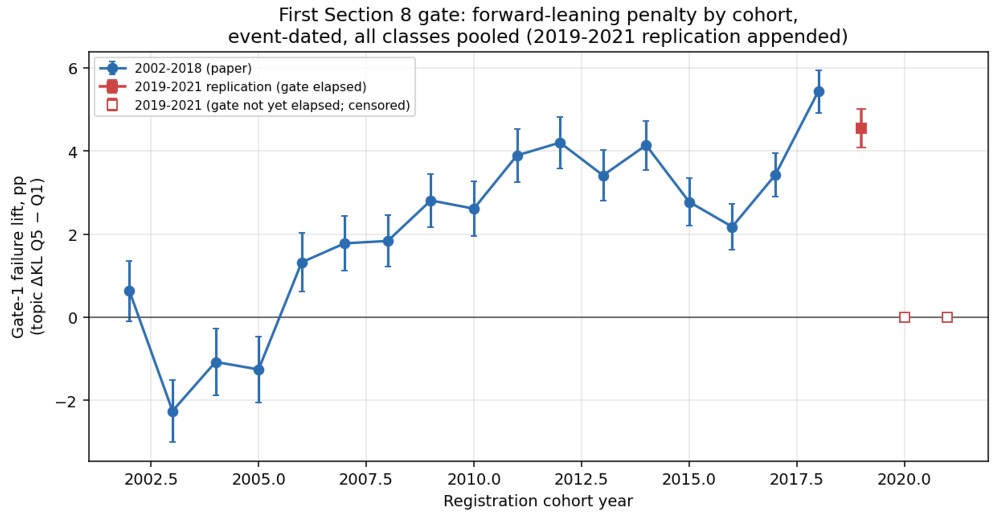

Figure 4: The leading gate penalty by registration cohort, all classes pooled. Each point is the event-dated first-gate failure rate of the most leading fifth of that cohort's registrations minus that of the most lagging fifth, with lead quintiles cut within the cohort and no controls. The penalty is absent or negative for the 2002–2005 cohorts, turns positive in 2006, and is present in every cohort from 2008 through 2019, ranging +1.8 to +5.6pp with no downward trend. 2019 is the last evaluable cohort: cancellations post at about age six and a half and the record ends in April 2026, so later cohorts have not reached their gate rather than having passed it.

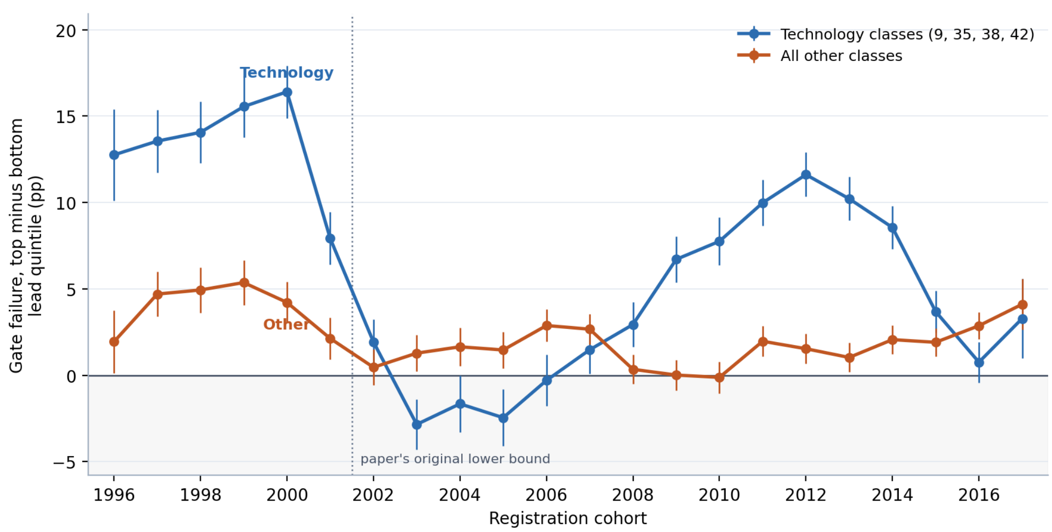

Figure 5: The leading gate penalty by registration cohort, technology classes (009, 035, 038, 042) against all others. Each point is the first-gate failure rate of the most leading fifth of that cohort's registrations minus that of the most lagging fifth, lead quintiles cut within cohort and group, no controls, with 95% intervals. Cohorts are restricted to registrations whose ninth anniversary is inside the record, so every outcome shown is settled. Outside technology the penalty is small and almost flat across two decades; inside technology it swings from +16pp for the 2000 cohort to −3pp for 2003 and back to +12pp for 2012. The dotted line marks the 2002 cut that the main cohort series uses.

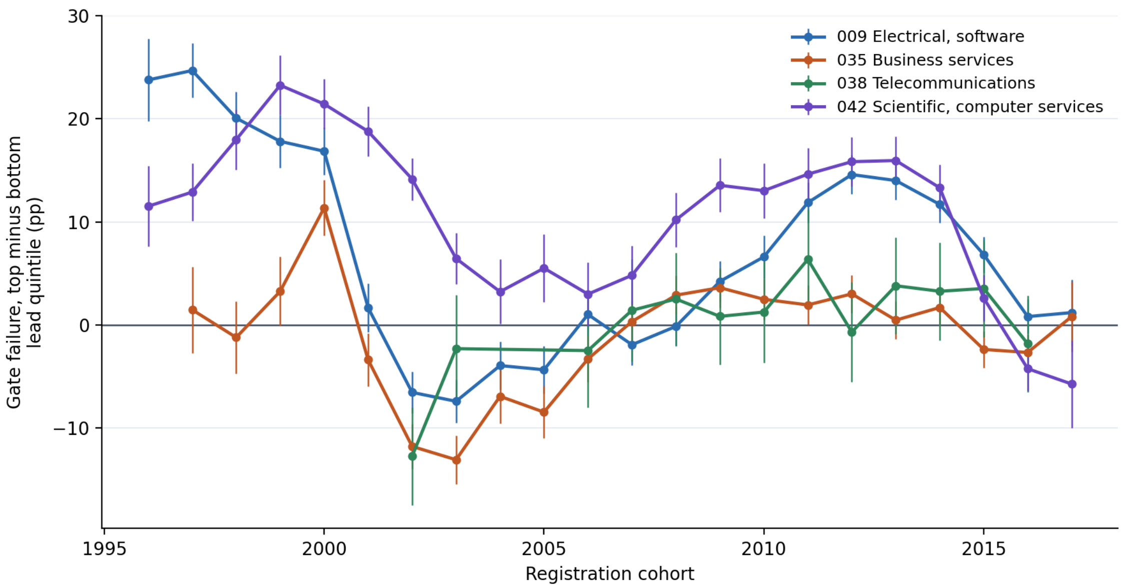

Figure 6: The leading gate penalty by registration cohort, one line per technology class. Each point is the first-gate failure rate of the most leading fifth of that class and cohort's registrations minus that of the most lagging fifth — lead quintiles cut within class and cohort, no controls — with 95% intervals, on registrations whose gate outcome is settled. Three classes cross zero between 2001 and 2006; scientific and computer services does not.

Table 6: Internet-bearing filings against the rest of their Nice class. Registration is the share of class-records filed 1995–2018 that reached registration; gate failure is the share of unique registrations 2002–2018 cancelled for non-use at age 4.0–8.5 years. Classes with fewer than 500 internet-bearing registrations are omitted.

Class		Web share	Registration (%)		Gate failure (%)	
		(%)	web	rest	web	rest
001	Chemicals	1.8	65.0	64.3	46.1	39.1
003	Cosmetics & Cleaning	2.8	58.0	54.2	52.1	54.4
004	Lubricants & Fuels	4.6	63.8	56.3	50.9	47.9
005	Pharmaceuticals	1.9	53.3	48.6	52.4	53.5
006	Metal Goods	3.5	68.1	67.0	44.5	39.1
007	Machinery	2.8	73.3	69.7	50.9	38.7
008	Hand Tools	3.6	65.8	62.5	42.4	43.6
009	Software & Electronics	18.7	53.3	57.0	56.5	51.4
010	Medical Apparatus	3.0	59.8	58.4	51.7	45.4
011	Lighting & Heating	3.0	67.4	64.8	51.4	45.9
012	Vehicles	3.4	69.0	60.9	49.6	45.1
014	Jewelry	7.4	63.2	55.8	54.7	55.7
016	Paper & Printed Goods	15.1	57.8	56.1	47.9	50.7
017	Rubber & Plastics	2.2	70.7	67.6	44.1	38.5
018	Leather Goods	8.0	61.4	57.4	50.5	55.4
019	Building Materials	2.0	71.4	64.0	47.5	43.9
020	Furniture	4.2	62.2	59.2	47.0	48.7
021	Household Utensils	5.1	60.1	59.0	44.5	50.5
024	Textiles	6.0	62.8	55.9	49.7	51.2
025	Clothing & Footwear	5.6	56.1	49.6	52.4	58.7
026	Lace & Embroidery	7.6	61.2	55.7	52.0	53.9
027	Carpets	5.6	66.8	56.8	43.7	46.9
028	Games & Sporting Goods	6.3	55.1	53.6	54.1	55.7
029	Meats & Processed Foods	2.5	59.7	55.7	53.3	49.6
030	Staple Foods	2.6	61.4	56.0	52.3	51.8
031	Agricultural Products	2.5	60.9	58.0	49.5	43.8
032	Beer & Soft Drinks	2.4	58.7	49.1	56.2	53.3
033	Alcoholic Beverages	1.2	61.3	52.7	56.9	42.3
034	Tobacco	2.9	53.8	49.1	55.0	54.2
035	Advertising & Retail	40.7	59.3	61.2	57.9	52.0
036	Insurance & Finance	18.9	58.6	62.6	52.4	48.9
037	Construction & Repair	10.4	67.8	67.5	49.0	47.1
038	Telecommunications	57.6	54.5	45.1	60.8	60.1
039	Transport & Storage	21.3	62.5	63.1	54.8	47.6
040	Material Treatment	9.6	68.0	63.9	49.4	47.3
041	Education & Entertainment	35.0	61.1	60.6	54.2	52.5
042	Scientific & Tech Services	39.3	57.8	59.4	58.0	51.4
043	Hotels & Restaurants	9.1	62.9	58.7	53.8	48.1
044	Medical & Beauty Services	20.6	62.9	62.1	53.7	51.3
045	Legal & Personal Services	41.5	60.3	65.0	60.1	48.7

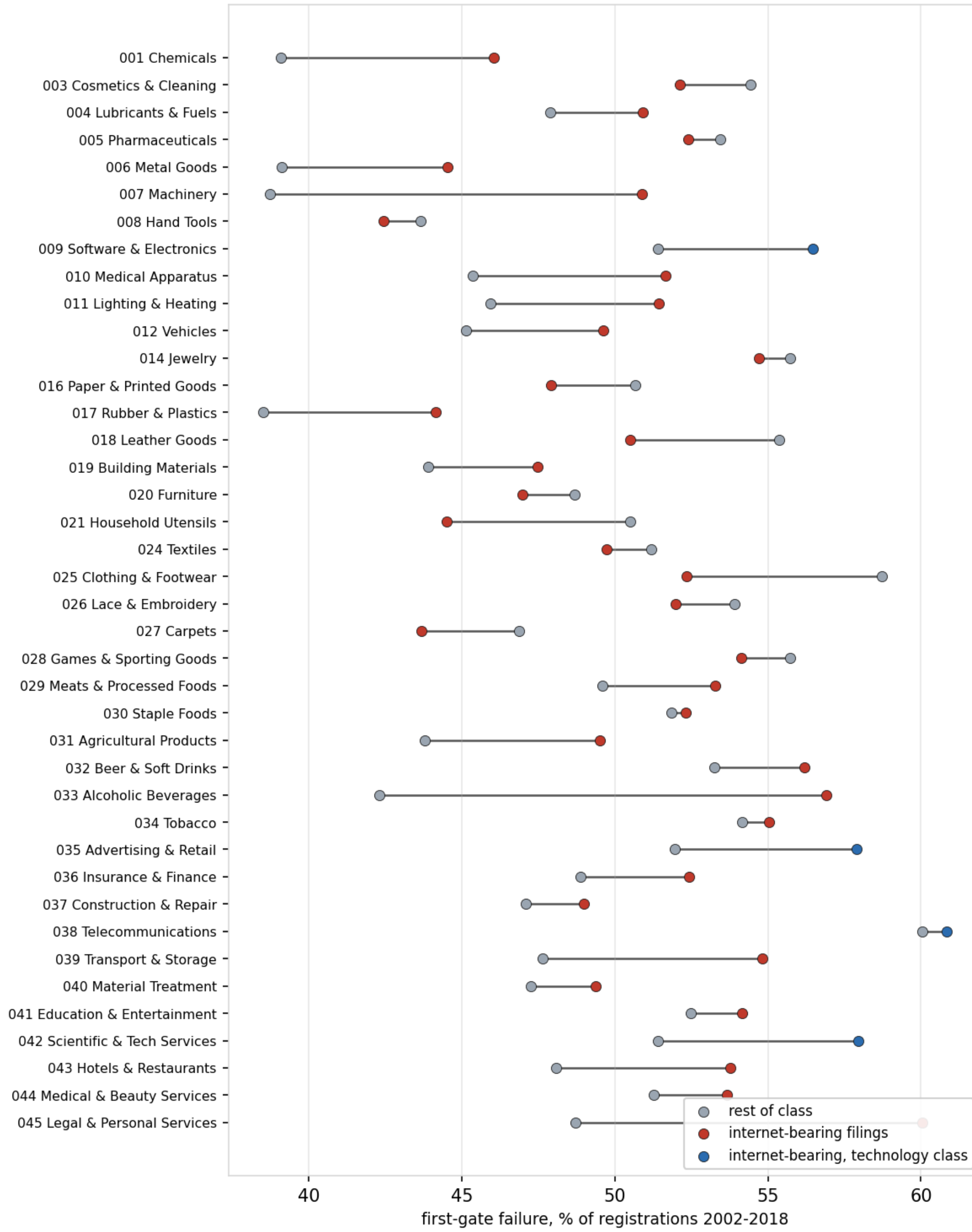

Figure 7: First-gate failure of internet-bearing registrations (red; blue in the four technology classes) against the rest of the same class (grey), registrations 2002–2018.

Table 7: First-gate failure contrasts under three theme scorings, on the identical registrations. Top-minus-bottom quintile, quintiles cut within Nice class and registration year, no controls; $n = 3{,}105{,}048$, base failure 52.3%.

Scoring	Lead Q5−Q1 (pp)	SE	Atypicality Q5−Q1 (pp)	SE
Global, 50 themes	+2.29	0.09	−4.39	0.09
Global, 500 themes	+2.61	0.09	−4.29	0.09
Per class, 50 themes	+1.94	0.09	−0.72	0.09

Figure 8: The internet gap in first-gate failure, inside each class. *Left:* internet-bearing minus other registrations' failure rate, by registration cohort in years since the internet pattern reached 2% of the class's filings, averaged across classes (band: one standard deviation across classes). *Right:* five classes in calendar time. Cells with fewer than 100 registrations in either group are dropped.

Table 8: The ladder by fifth of debut lead and of debut atypicality. Rates in percent of owners whose first registered filing was made 2009–2018 ($n = 999{,}442$); fifths cut within Nice class, debut year, and fifth of description length. The last column is the most-minus-least contrast with its t-statistic.

Rung	Axis	Q1	Q2	Q3	Q4	Q5	Q5−Q1 (pp; t)
Funded	lead	1.86	1.73	1.61	1.57	1.80	−0.06 (−1.4)
	atypicality	1.68	1.70	1.79	1.76	1.64	−0.04 (−1.1)
Reporting	lead	0.59	0.49	0.40	0.34	0.38	−0.21 (−9.4)
	atypicality	0.42	0.36	0.43	0.46	0.53	+0.11 (+5.2)
IPO	lead	0.24	0.21	0.14	0.13	0.17	−0.07 (−4.5)
	atypicality	0.15	0.15	0.18	0.21	0.22	+0.07 (+4.8)

Table 9: Debut listing and vocabulary position: design-based estimates. LPM of P(owner ever in SEC EDGAR) on 1,556,968 registered debut filings 1995–2018, one observation per owner, class×filing-year fixed effects, heteroskedasticity-robust SEs. Quintiles cut within class×year. Base rate 0.481%.

	Coefficient (pp)	SE
Atypicality quintiles, $A = \frac{1}{2}(K^- + K^+)$ *(Q1 omitted)*		
Q5, FE only	+0.017	0.018
Q5, FE + log length	+0.000	0.018
Q5, FE + log length, on K^- alone	−0.003	0.018
Signed Δ*KL quintiles, FE + log length + KL levels*		
Q3	−0.133	0.020
Q4	−0.187	0.021
Q5	−0.158	0.024
Decomposition, FE + log length		
$\lvert\Delta\text{KL}\rvert$ (pp per sd)	+0.067	0.009
signed ΔKL (pp per sd)	−0.047	0.006
log description length	+0.056	0.004

Standard errors in place of significance stars; see the note to Table 3.

Table 10: Every outcome in the funnel, conditioned on its prerequisite, under one specification. Unit is the debut owner, scored on that owner's first filing. Each row is a linear probability model with class×debut-year fixed effects and heteroskedasticity-robust errors; coefficients are percentage points per standard deviation. *Atypicality* is the average of the two KL levels; *lead* is the signed difference. Cohort windows differ because each maintenance gate needs elapsed time and Form D is electronic only from 2009, so the rows are not a nested decomposition. The first gate is a dated cancellation for non-use; the second is whether a registration that cleared the first was renewed rather than cancelled or expired, restricted to 2002–2013 cohorts, since a year-six and a year-ten death share a terminal status and are separable only this way. This table alone is estimated at $T = 200$ topics rather than the $T = 50$ used elsewhere; the sign and ordering of every row are unchanged at $T = 50$.

			Unsigned		Signed
Outcome *(population)*	n	Base	Atyp.	$\lvert\Delta\text{KL}\rvert$	Lead
Reached registration *(filed)*	3,161,552	56.11%	−1.352 (0.033)	−0.307 (0.044)	+0.872 (0.028)
Passed gate 1, yr 6 *(registered)*	1,260,392	41.82%	+1.349 (0.054)	−1.150 (0.082)	−0.487 (0.052)
Passed gate 2, yr 10 *(gate 1)*	297,944	55.92%	+1.296 (0.108)	−0.297 (0.162)	−0.629 (0.105)
Raised Reg D round *(filed)*	1,535,004	2.53%	+0.003 (0.015)	+0.148 (0.027)	−0.149 (0.016)
Raised Reg D round *(registered)*	916,448	2.61%	−0.017 (0.020)	+0.115 (0.035)	−0.114 (0.021)
Listed after round *(funded)*	38,886	3.49%	+0.493 (0.129)	+0.608 (0.170)	−0.513 (0.104)
Became SEC-reporting *(registered)*	1,773,938	0.56%	+0.042 (0.007)	+0.040 (0.011)	−0.004 (0.007)

Standard errors in parentheses, in place of significance stars; see the note to Table 3.

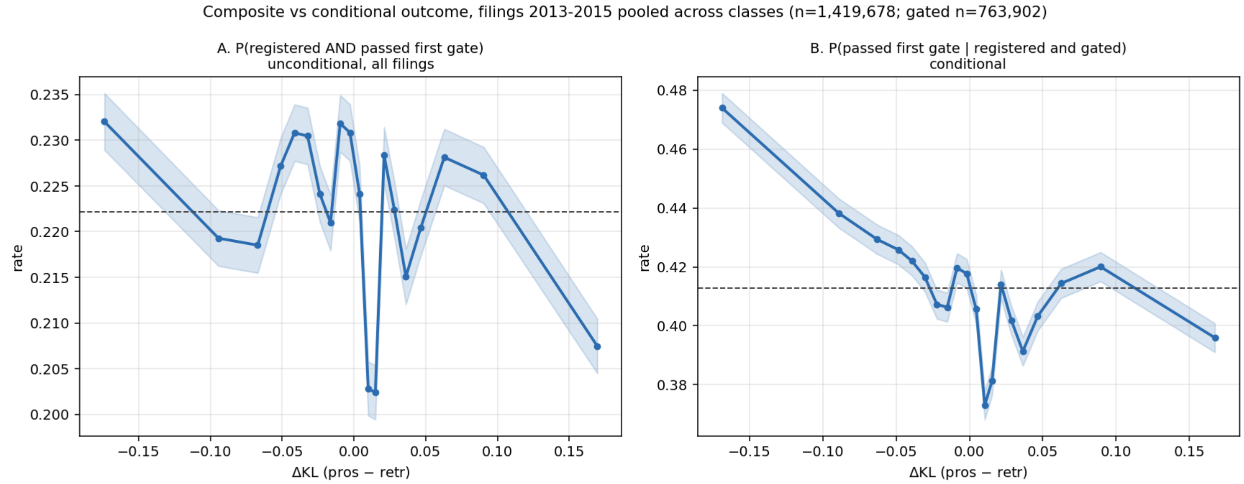

Figure 9: Filings 2013–2015 pooled across classes, theme-scored on per-filing reference windows (n = 1,419,678; gated subset n = 763,902). *A:* the unconditional composite P(registered and passed first gate) over all filings. *B:* P(passed | registered and gated) on the same filings. The unconditional view is nearly flat in lead (22.4% to 22.1% across quintiles); conditioning on grant is what exposes the gradient.

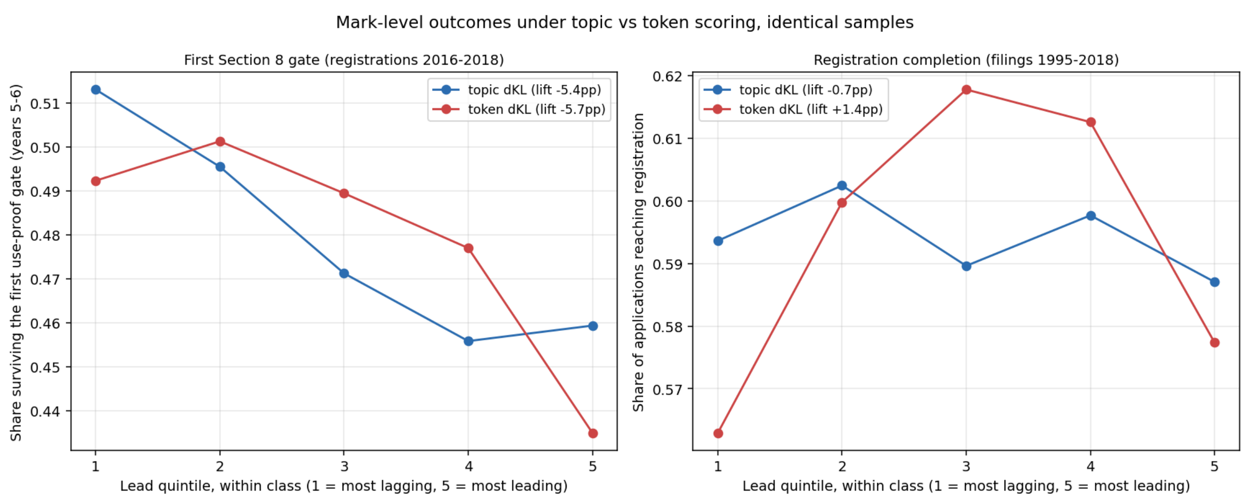

Figure 10: The two mark-level outcomes under theme and term scoring on identical samples, pooled across classes, by within-class lead quintile (1 most lagging, 5 most leading). *Left:* the share of registrations 2016–2018 that survived the first use-proof gate at years five to six, read from terminal status codes, where the two representations agree closely. *Right:* registration completion (filings 1995–2018), where they do not: the inverse-U in the signed axis is present under term scoring (depth +0.048) and absent under theme scoring (depth −0.001). The gate result is scoring-robust; the registration curvature on this axis is not, and Section 3 reports only the unsigned version.

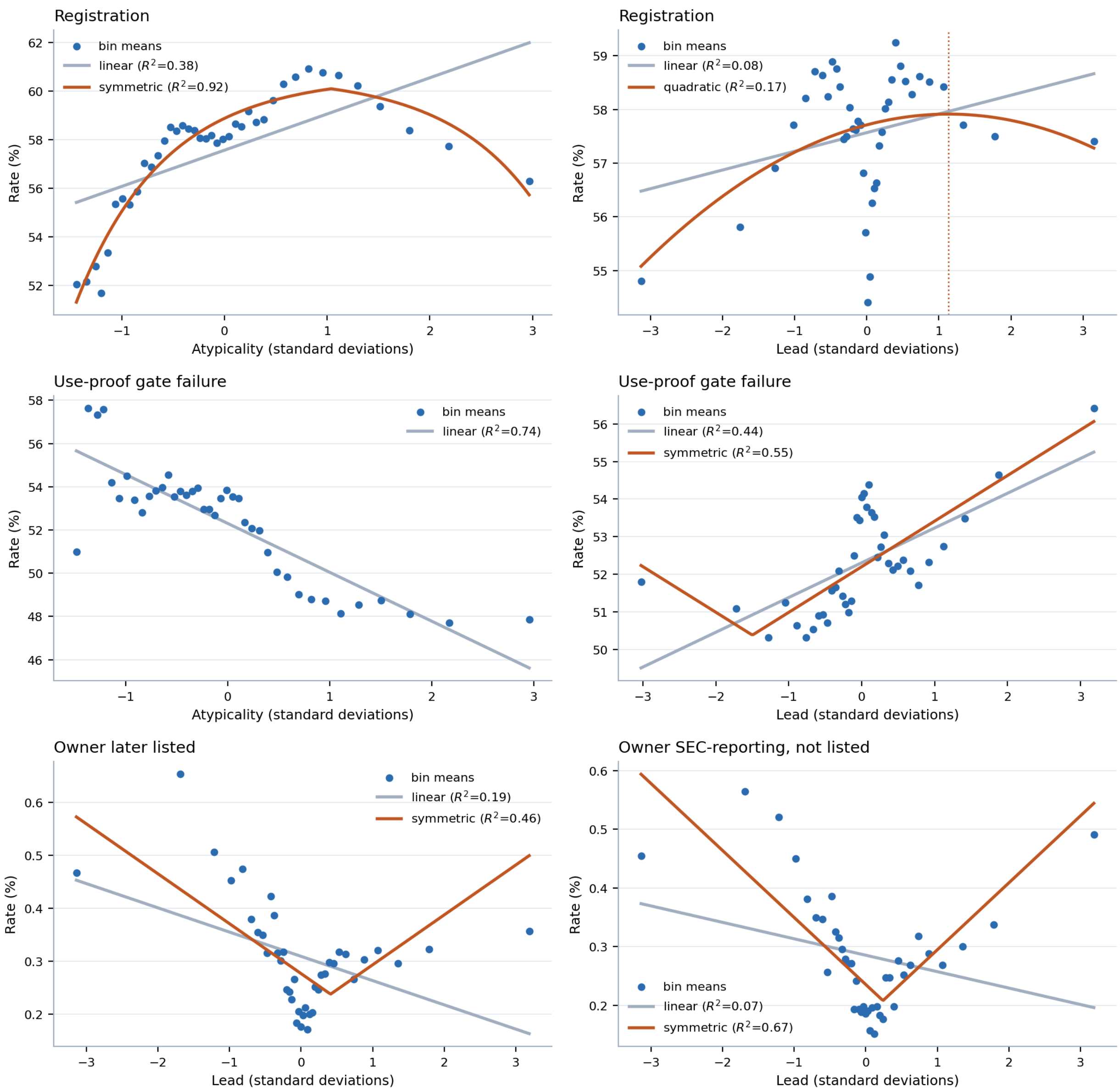

Figure 11: Binned profiles at the three gates, with the linear reading the paper's contrasts assume and the shape selected by AIC where that is not linear. Forty equal-count bins; the dotted line marks the fitted vertex. The bottom row is the case that matters: a linear fit through the public-reporting profile slopes downward while the data turn up at both tails.

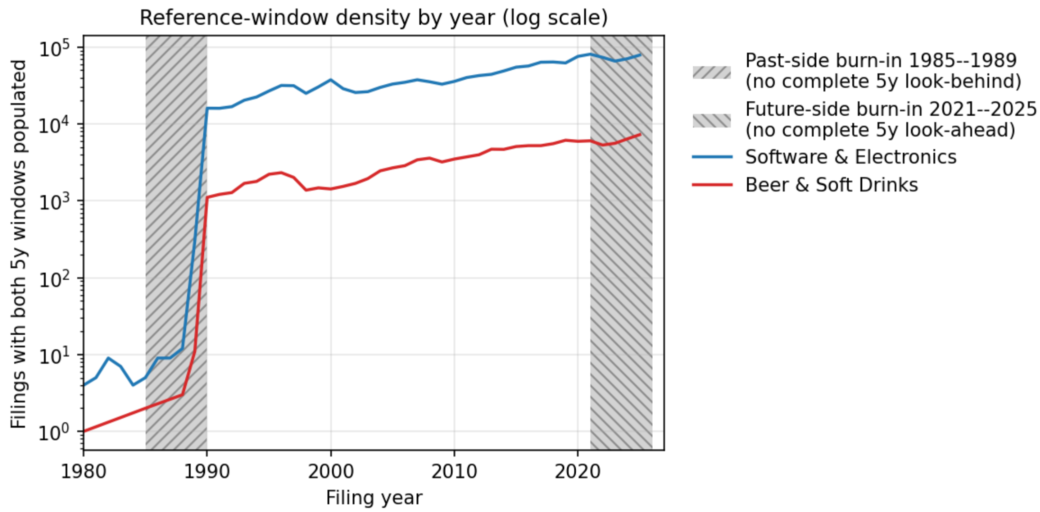

Figure 12: Reference-window density by year on log scale. Hatched regions are the past-side and future-side burn-in zones where the 5-year window is incomplete. The windows determine only how surprising a filing's language is — its score at the filing date. The outcome the score is tested against is read separately and later: whether the registration survived the use-proof gate, decided at registration age six to seven, which is why the score is interpretable from 1995 but the gate sample ends with the 2018 cohort.

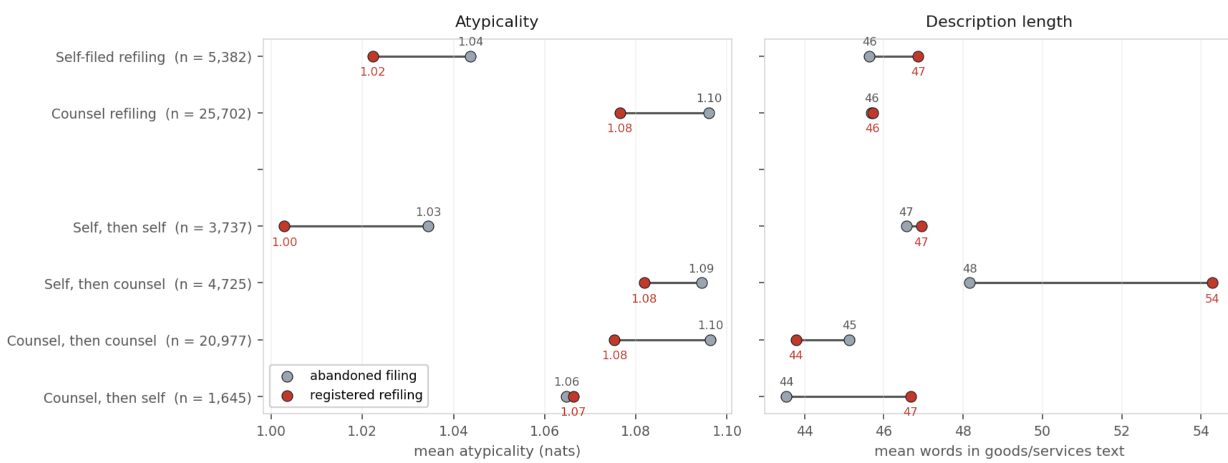

Figure 13: Rewritten refilings: mean atypicality and mean description length of the abandoned filing (grey) and of the registered refiling (red), by representation on the refiling and by the change in representation between the two attempts. Mean atypicality falls by 0.01 to 0.03 nats in every group except counsel-then-self, where it is unchanged; the means are small because the convergence runs from both tails, which the profile by fifth below shows. Length rises by six words where counsel is hired for the second attempt and falls by one where counsel rewrites counsel.

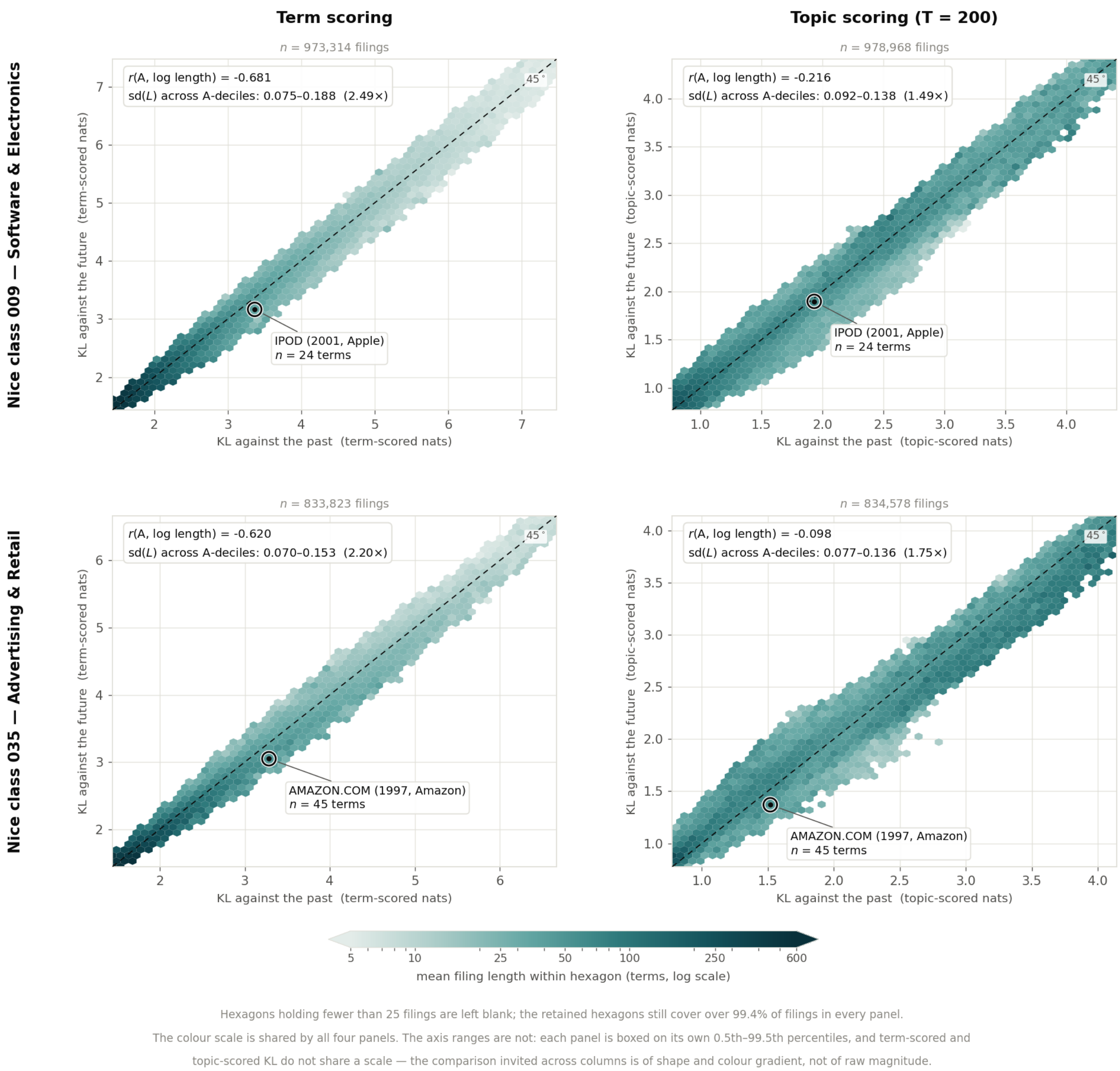

Figure 14: Mean filing length across the two axes, by representation and class, for filings 1995–2019 scored on per-filing reference windows. The horizontal axis is surprise against the preceding five years, the vertical axis surprise against the following five; a filing on the dashed diagonal is equally far from both, and distance below it is lead. Hexagons holding fewer than 25 filings are blank; the retained hexagons cover over 99.4% of filings in every panel. The colour scale is shared across panels; the axis ranges are not, since term-scored and theme-scored KL do not share a scale, so the comparison invited across columns is of shape and colour gradient rather than of magnitude. Both columns are drawn on the filings the per-filing windows score, which differ between representations by under one percent.

Table 11: Top theme loading for three filings, by representation. Term scoring resolves the novel content of all three; theme scoring resolves none of it at any tested resolution.

	Amazon.com (1997)	Kombucha (1997)	Ethereum (2018)
novel content (term)	line search, search ordering	kombucha, saccharose	blockchains, distributed computing
$T = 50$	video, computer, music (0.40)	beverages, fruit, drinks (0.64)	software, online, downloadable (0.33)
$T = 200$	entertainment, video, games (0.29)	beverages, drinks, coffee (0.60)	field, services, development (0.37)
$T = 500$	computer, software, data (0.22)	beverages, drinks, fruit (0.61)	software, online, computer (0.49)